

Heat literature, original source metrics, and the observed-class return

Split-Zero research programme

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The original observation quotient now receives the terminal class with at most $O_{h,\varpi}(k \log q)$ logarithmic loss throughout the complete window $q - 1 \leq N \leq 2q$, $q = (k + 1)^2$, on an explicitly bounded original large-period domain. This evaluates its four-cutoff norm return and degree-resolved loss profile through every term of size $q/\log q$. The full phase and corner calculation is included. We also sharpen the same-class arithmetic–Gamma comparison to $O_h(k + \log q)$ and improve an earlier conductor upper bound. These estimates concern specified original minima; they do not assign the independent kernel allocation, target minimum or projected-current phases.

The accompanying literature calculation uses the author LaTeX of Burnol, Connes, Connes–Consani–Moscovici and Polymath. It proves their exact maps into the existing source, its full jets, Gamma factors and attained quotients. The Gaussian metric evolution retains its moving minimizer, and the tensor calculation retains the distinction between factorwise and sum-coordinate heat through their explicit algebra maps. All earlier finished estimates are preserved; the input audit is checked at its stated excerpt scope.

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1 What the incoming conductor audit establishes

The supplied audit explicitly reports an incomplete reading of two source excerpts. Its equations (1)–(17) are checked here against the complete *Weighted conductor forward* source, WCF1–27. It is evidence for those finite calculations, not evidence that the unread programme lacks their successors. In particular WCF19a–f already proves the full determinant and inverse-exterior argument, WCF22 gives the actual lower evaluation map, and WCF27 explicitly retains the error at the $q \log a$ scale. The audit’s full-determinant comparison is compatible with these statements. We now prove a stronger finite bound using its sharper weight estimate and the compression argument in WCF11a–12.

Fix $a = 4\ell + 1 \geq 9$, $q = (a+1)^2$, $q' = (a-7)^2$, the original $0 < \delta < 1/2$, $\gamma > 2$ and coefficients a_{uw} , $0 \leq u, w \leq 8$. Put

$$E_A(z) = \sum_{u,w=0}^8 a_{uw} e^{[4+(2u-8)\delta+i(2w-8)\gamma]z}, \quad v = \text{ord}_0 E_A, \quad \mu_v = E_A^{(v)}(0) \neq 0, \quad A_{\text{abs}} = \sum_{u,w} |a_{uw}|. \quad (\text{AC1})$$

The nonzero symbol is fixed as a varies; $0 \leq v \leq 80$. For $s \in \{1, a\}$, $b = s/2$, $N = D + v$, and $D \geq 0$, retain

$$\rho_n^2 = \frac{n!}{(b)_n}, \quad w(t) = \frac{1}{2} \log \frac{1-it}{1+it}, \quad g(t) = t^{-v} e^{-4w(t)} E_A(w(t)), \quad (\text{AC2})$$

$$A_{D,s} = \text{diag}(\rho_i)_{0 \leq i \leq D}^{-1} T_D(g) \text{diag}(\rho_{j+v})_{0 \leq j \leq D}.$$

The logarithm is the branch vanishing at zero. The two original polynomial spaces have the WCF2 measure

$$dm_s(y) = \frac{(2\pi)^{s/2} 2^{s/2}}{4\pi \Gamma(s/2)} |\Gamma(s/4 + iy/2)|^2 dy$$

and centres $a/2$ and $a/2 - 4$. No measure or leading symbol coefficient is changed. Their orthonormal coordinates give (AC2); see WCF4–6, whose derivation uses the Meixner–Pollaczek generating function [7, 18.23.7].

Proposition 1 (Sharpened original forward bound). *For $D \geq 1$ define*

$$\widehat{C}_N = e^{1+2\pi\gamma} A_{\text{abs}} (2N+1)^{4\delta}, \quad \widehat{B}_{D,s} = \begin{cases} \widehat{C}_N (4N+1)^{1/4}, & s = 1, \\ \widehat{C}_N, & s = a. \end{cases} \quad (\text{AC3})$$

Then $\|A_{D,s}\| \leq \widehat{B}_{D,s}$. At $D = 0$ its exact norm is $|\mu_v| \rho_v / v!$. Every constant refers to the original symbol.

Proof. For $|t| = r < 1$, the Cayley ratio lies in the right half-plane and has modulus between $(1-r)/(1+r)$ and its reciprocal. Each exponent of that ratio in $e^{-4w} E_A(w)$ has real part bounded by 4δ and imaginary part bounded by 4γ . Hence

$$\max_{|t|=r} |g(t)| \leq r^{-v} A_{\text{abs}} e^{2\pi\gamma} ((1+r)/(1-r))^{4\delta}. \quad (\text{AC4})$$

For $b > 1$, the exact coefficient norm is

$$\sum |c_n|^2 \rho_n^2 = (b-1) \int_0^1 \int_0^{2\pi} \left| \sum c_n x^{n/2} e^{in\theta} \right|^2 (1-x)^{b-2} \frac{d\theta}{2\pi} dx.$$

Angular orthogonality and Euler's beta integral prove this identity [6, 5.12.1]. Multiplication by $g(rz)$ followed by projection onto degrees at most D therefore has norm at most $\max_{|t|=r} |g(t)|$. Its transpose in the basis z^n/ρ_n is $\text{diag}(\rho_i)^{-1} T_D(g(\cdot)) \text{diag}(\rho_i)$. Conjugating by $\text{diag}(1, r, \dots, r^D)$ costs r^{-D} . The remaining diagonal ρ_{i+v}/ρ_i is a contraction. Thus $\|A_{D,a}\| \leq r^{-D} \max |g|$.

For $b = 1/2$, use circle Parseval in place of the beta norm. The same compression gives $\|T_D(g)\| \leq r^{-D} \max |g|$ and the original diagonal factors cost ρ_N , since ρ_n increases from one. We prove $\rho_n \leq (4n+1)^{1/4}$ by induction. It holds at zero; the squared induction step follows from

$$(4n+5)(n+\frac{1}{2})^2 - (4n+1)(n+1)^2 = \frac{1}{4} > 0.$$

Choose $r = N/(N+1)$, which is allowed because $D \geq 1$. Now $r^{-D-v} = r^{-N} \leq e$ and $(1+r)/(1-r) = 2N+1$. Combining these facts with (AC4) proves (AC3). For $D = 0$, (AC2) is the scalar $(-i)^v \mu_v \rho_v / v!$. $\square$

Proposition 2 (Determinant and original four-endpoint receiver). *Put $d = D + 1$. The exact complex determinant and its two-source comparison are*

$$\det A_{D,s} = ((-i)^v \mu_v / v!)^d \prod_{j=0}^D \rho_{j+v} / \rho_j, \quad |\det A_{D,s}|^2 = |\mu_v / v!|^{2d} \prod_{j=0}^D \prod_{r=1}^v \frac{j+r}{j+b+r-1}, \quad (\text{AC5})$$

$$0 \leq \log |\det A_{D,1}| - \log |\det A_{D,a}| \leq \frac{v(a-1)}{2} H_{D+1}. \quad (\text{AC6})$$

For the actual WCF lower evaluation matrices $E_i = \mathbf{E}_{-,D_i,s}$, $N_i = (q-1, q, 2q-1, 2q)_i$, $D_i = N_i - v$, define

$$e_i = \log \frac{\det(E_i^* \bar{A}_{D_i,s} A_{D_i,s}^T E_i)}{\det(E_i^* E_i)}, \quad \mathcal{R}e = -e_0 - e_1 + e_2 + e_3.$$

Set $M_i = \max(1, \hat{B}_{D_i,s})$ and

$$J_i = (D_i + 1) \log^+ |v!/\mu_v| + \frac{1}{2}(s/2 - 1)_+ v H_{D_i+1}, \quad m = 16a - 48 - v.$$

Then the original conductor correction satisfies

$$\mathcal{R}e \leq 2m \log M_0 + 2(m+1) \log M_1 + 2J_0 + 2J_1 + 2q' \log M_2 + 2q' \log M_3. \quad (\text{AC7})$$

For the fixed actual symbol, this improves WCF26a to

$$\mathcal{R}e \leq \begin{cases} (32\delta + 2)q \log a + O_{\text{actual}}(q), & s = 1, \\ 32\delta q \log a + O_{\text{actual}}(q), & s = a. \end{cases} \quad (\text{AC8})$$

This is an upper bound, not an assigned endpoint value.

Proof. Triangularity proves (AC5). Dividing its two squared moduli gives one half the sum of $\log(1 + (a-1)/(2(j+r-1/2)))$. Use $\log(1+x) \leq x$ and $j+r-1/2 \geq (j+1)/2$ to prove (AC6). The gamma-product version in the audit follows by multiplying the consecutive factors in j . For fixed v and $D = O(a^2)$, (AC6) divided by q tends to zero. This comparison is at the same dimension and concerns the full matrix.

For any invertible $d \times d$ matrix B , its singular values give $\|\bigwedge^r B^{-1}\| = \|\bigwedge^{d-r} B\|/|\det B|$. Formula (AC5) and $\log(1+x) \leq x$ give $|\det A_{D_i,s}|^{-1} \leq e^{J_i}$. Consequently the determinant of the Gram of any q' independent vectors after applying $A_{D_i,s}^T$ lies between its original determinant times $e^{-2((D_i+1-q') \log M_i + J_i)}$ and times $M_i^{2q'}$. This follows by applying the exterior map, then its inverse, to their full wedge. WCF22 identifies these vectors with the actual conductor evaluation columns; no arbitrary endpoint substitutes for them. At the low endpoints $D_0 + 1 - q' = m$ and

$D_1 + 1 - q' = m + 1$. Insert the lower bounds at those two endpoints and upper bounds at the two high endpoints to obtain (AC7), with its stated signs.

For the high endpoints $N_i = 2(a + 1)^2 + O(1)$, (AC3) gives $\log M_i \leq 8\delta \log a + O_{\text{actual}}(1)$ at $s = a$, and $(8\delta + 1/2) \log a + O_{\text{actual}}(1)$ at $s = 1$. The low-dimensional first two terms in (AC7) are $O(a \log a)$; $J_0 + J_1 = O_{\text{actual}}(q + a \log a)$. Since $q' \leq q$, (AC8) follows. The original WCF27 complete action receives $\mathcal{R}e$ with coefficient one; replace its corresponding upper allowance by the minimum of the old allowance and (AC7). All other action terms remain. $\square$

The audit's trace-log remainder follows by diagonalizing its Hermitian H and bounding the scalar series remainder by $|\lambda|^{M+1}/((M+1)(1-\theta))$. Summation yields both its rank and trace-norm versions. Its Vandermonde estimate follows from $\sum_n |[z^n] \prod_{k \neq j} (z - \beta_k)| \leq (1+B)^{80}$ and $\prod_{k \neq j} |\beta_j - \beta_k| \geq (2\delta)^{80}$; it bounds the largest of the first 81 moments, not necessarily the first nonzero one. These are valid checks, with the stated scopes. Neither implies individual projected-current signs or absolute native allocations.

2 Original source, Burnol's division functions and local factors

Let $D = -x\partial_x$, $\mathcal{H} = L^2(\mathbb{R}_{>0}, dx)$ and

$$g(s) = 2\xi(s) = s(s-1)L_\infty(s)\zeta(s), \quad L_\infty(s) = \pi^{-s/2}\Gamma(s/2), \quad h(s) = \prod_{\rho \in Z} (s - \rho)^{m_\rho}, \quad d = \deg h. \quad (\text{LR1})$$

Here Z is a fixed finite set of nontrivial zeros and each m_ρ is its full multiplicity. The original source a_h has Mellin transform g/h . Write $\mathcal{M}f(s) = \int_0^\infty f(x)x^{s-1}dx$, $j_h F = (F^{(r)}(\rho)/r!)_{\rho, r < m_\rho}$, and $\mathsf{T}_h = j_h(g/h) \cdot$, multiplication in the full jet algebra. It is invertible. The physical source and its jet are exactly

$$\begin{aligned} \mathcal{A}_N P &= P(D)a_h, \quad j_h \mathcal{M}(\mathcal{A}_N P) = \mathsf{T}_h j_h P, \\ \|\mathcal{A}_N P\|^2 &= \int_{\mathbb{R}} |P(1/2 + iy)|^2 w_h(y) dy, \quad w_h(y) = \frac{|(g/h)(1/2 + iy)|^2}{2\pi}. \end{aligned} \quad (\text{LR2})$$

Every formula below uses this measure including its 2π factor.

Proposition 3 (Burnol-to-original-source map, with full divided jets). *Let P have degree below d . Its exact partial fraction decomposition is*

$$\frac{P(s)}{h(s)} = \sum_{\rho \in Z} \sum_{l=1}^{m_\rho} \frac{c_{\rho l}(P)}{(s - \rho)^l}, \quad c_{\rho l}(P) = \frac{1}{(m_\rho - l)!} \left[\partial_s^{m_\rho - l} \frac{P(s)}{h(s)/(s - \rho)^{m_\rho}} \right]_{s=\rho}. \quad (\text{LR3})$$

There is a unique $b_P \in L^2(0, \infty; dx)$ with right Mellin transform $\mathcal{M}_R b_P(s) = \zeta(s)P(s)/h(s)$. It belongs to Burnol's L_1 (both it and its cosine transform are constant on $(0, 1)$). Define $I f(x) = x^{-1}f(1/x)$ and the bounded Mellin multiplier $T_m = \mathcal{M}^{-1}m(s)\mathcal{M}$, $m(s) = s(s-1)L_\infty(s)$, on the line $\Re s = 1/2$. Then

$$\mathcal{A}_{d-1} P = T_m I b_P, \quad \|\mathcal{A}_{d-1} P\|^2 = \int_{\mathbb{R}} |m(1/2 + iy)|^2 |\mathcal{M}_R b_P(1/2 + iy)|^2 \frac{dy}{2\pi}. \quad (\text{LR4})$$

Thus the original canonical section is obtained by inserting $P = r_h(u)$, $j_h r_h(u) = \mathsf{T}_h^{-1}u$ in (LR3)–(LR4).

Proof. Principal parts give (LR3); after subtracting them the rational function has no poles and vanishes at infinity, so it is zero. Burnol [1, §4, Proposition **zetaprop** and its proof] proves that each $\zeta(s)/(s - \rho)^l$, $l \leq m_\rho$, is the right Mellin transform of a vector of L_1 . Its proof uses the exact Hardy-space characterization of L_1 and $\chi(s)\zeta(1-s)/(1-s-\rho)^l = (-1)^l \zeta(s)/(s - (1-\rho))^l$; it

does not assume RH. The finite linear combination (LR3) therefore defines b_P . Mellin Plancherel proves uniqueness. Substitution $t = 1/x$ shows both $\|If\| = \|f\|$ and $\mathcal{M}(If)(s) = \mathcal{M}_R f(s)$. The function m is continuous on this line and bounded: the Gamma asymptotic [6, 5.11.9] gives polynomial growth times $e^{-\pi|y|/4}$ at either end. Consequently T_m is bounded. Its product with $\zeta P/h$ is exactly gP/h , proving (LR4) by Plancherel. The multiplier has no zero on the line, so the map is injective. Its inverse on this finite-dimensional image is the restriction of multiplication by $1/m$; no unweighted isometry is asserted. $\square$

For the full original polynomial space, division gives a unique $P = r + hQ$, $\deg r < d$. Consequently

$$\mathcal{A}_N P = T_m I b_r + Q(D) f_0, \quad \mathcal{M} f_0 = g, \quad h(D) a_h = f_0. \quad (\text{LR5})$$

This explicitly retains every relation vector and its cross terms in the norm. It does not assert that ζQ is an L^2 Mellin transform. The full quotient minimum below includes the complete second summand.

Proposition 4 (Exact semilocal moment receiver). *For a finite set S of places containing ∞ , put*

$$\begin{aligned} a_S(y) &= L_\infty(s) \prod_{p \in S \setminus \{\infty\}} (1 - p^{-s})^{-1}, \quad s = 1/2 + iy, \quad d\mu_S(y) = |a_S(y)|^2 dy, \\ q_{S,h}(y) &= \frac{g(s)}{\sqrt{2\pi} h(s) a_S(y)} = \frac{s(s-1)\zeta(s)}{\sqrt{2\pi} h(s)} \prod_{p \in S \setminus \{\infty\}} (1 - p^{-s}). \end{aligned} \quad (\text{LR6})$$

Multiplication $U_{S,h} F = q_{S,h} F$ is unitary from $L^2(w_h dy)$ to $L^2(d\mu_S)$. It intertwines multiplication by y on their full maximal domains. The native polynomial space maps exactly to $q_{S,h} \mathcal{P}_N$, and its norm, restrictions, full relation minima and attained observation quotients are preserved under this map.

Proof. The density identity $|q_{S,h}|^2 |a_S|^2 = w_h$ proves isometry. The finite Euler factors have no zero on the line. The quotient g/h is entire and its zeros there are discrete, so $q_{S,h} \neq 0$ almost everywhere. For any $V \in L^2(d\mu_S)$, the function $V/q_{S,h}$ has squared w_h norm equal to $\|V\|^2$, proving surjectivity. The same identity with $|yF|^2$ proves the domain and intertwining assertion. Applying this identity to each vector in every original affine fibre preserves its infimum and its minimizing vector. It also preserves each cross inner product by polarization. $\square$

Connes–Consani–Moscovici [2, §3.2, Proposition 3.1] prove determinacy for precisely this local-factor density (their choice of sign on the line leaves the modulus unchanged). Their Gamma estimate bounds its even moments by $M^n(2n)!$. No division by total mass has occurred in (LR6). To make its Jacobi form explicit, let $V_S : L^2(\mu_S) \rightarrow \ell^2$ be expansion in its orthonormal polynomials, and put $J_S = V_S M_y V_S^{-1}$ and $\xi_S = \sqrt{\mu_S(\mathbb{R})} e_0$. Polynomial completeness here follows, for example, directly from the exponential tail: if F is orthogonal to every polynomial, the Fourier transform of $F\mu_S$ is analytic in a strip by Cauchy–Schwarz, has every derivative zero at zero, and hence vanishes; Fourier uniqueness gives $F = 0$. Thus V_S is unitary. For the original polynomials P_i one has the exact full Gram

$$G_{ij} = \langle q_{S,h}(J_S) P_i (1/2 + iJ_S) \xi_S, q_{S,h}(J_S) P_j (1/2 + iJ_S) \xi_S \rangle. \quad (\text{LR7})$$

The functional-calculus vectors are in their stated domains because (LR2) is finite. The root polynomial, zeta factor and finite Euler factors in (LR6) remain inside this expression. In particular no finite Euler-product limit on the critical line is used.

3 Evaluating the original interpolation minimum by its kernel

All moments of w_h are finite by the Gamma decay in (LR1), and its support is infinite. Let p_n be its real monic orthogonal polynomials, $h_n = \int p_n(y)^2 w_h(y) dy > 0$. These are auxiliary coordinates in the unchanged source space, not a replacement of its measure. They can be computed without an unknown scalar: with $c_j = \int y^j w_h(y) dy$ and $\Delta_{n-1} = \det(c_{i+j})_{0 \leq i,j < n}$, $\Delta_{-1} = 1$,

$$p_n(z) = \frac{1}{\Delta_{n-1}} \det \begin{pmatrix} c_0 & c_1 & \cdots & c_n \\ c_1 & c_2 & \cdots & c_{n+1} \\ \vdots & \vdots & & \vdots \\ c_{n-1} & c_n & \cdots & c_{2n-1} \\ 1 & z & \cdots & z^n \end{pmatrix}, \quad h_n = \Delta_n / \Delta_{n-1}. \quad (\text{LR8})$$

Expansion of the determinant shows it is monic and orthogonal to the first n monomials; the norm identity follows by block elimination. This is the moment construction used in [2, §2.1], with all mass factors retained.

Proposition 5 (Full-jet Christoffel–Darboux minimum). *For $N \geq d - 1$ set*

$$K_N(z, w) = \sum_{n=0}^N \frac{p_n(z) \overline{p_n(w)}}{h_n} = \frac{p_{N+1}(z) \overline{p_N(w)} - p_N(z) \overline{p_{N+1}(w)}}{h_N(z - \bar{w})}. \quad (\text{LR9})$$

At $z = \bar{w}$ use the removable value and its derivatives. For each n , retain the complete physical-jet vector

$$t_n = j_h \left[(g/h)(s) \frac{p_n((s - 1/2)/i)}{\sqrt{h_n}} \right], \quad M_N = \sum_{n=0}^N t_n t_n^*. \quad (\text{LR10})$$

Then $M_N > 0$, and the original attained metric is

$$H_N = M_N^{-1}, \quad \min_{T_h j_h P = u} \|\mathcal{A}_N P\|^2 = u^* H_N u. \quad (\text{LR11})$$

Thus every matrix entry in M_N is obtained by taking the full divided derivatives in (LR9), including the g/h factors and i^{-r} factors from $s = 1/2 + iy$. The exact next-cutoff update is

$$H_{N+1} = H_N - \frac{H_N t_{N+1} t_{N+1}^* H_N}{1 + t_{N+1}^* H_N t_{N+1}}, \quad \log \frac{\det H_{N+1}}{\det H_N} = -\log(1 + t_{N+1}^* H_N t_{N+1}). \quad (\text{LR12})$$

Proof. Symmetry of multiplication by y gives $yp_n = p_{n+1} + \alpha_n p_n + (h_n/h_{n-1})p_{n-1}$; coefficients of all lower p_j vanish by orthogonality. Multiply this recurrence at z by $p_n(w)/h_n$, subtract its conjugate at w , and sum. All internal terms cancel, leaving (LR9). This proves the classical Christoffel–Darboux formula [7, 18.2.12] also at complex points by polynomial continuation.

Write $P(s) = \sum_{n=0}^N c_n p_n((s - 1/2)/i)/\sqrt{h_n}$. Then its exact squared source norm is $c^* c$ and its physical jet is $T_N c$, $T_N = (t_0, \dots, t_N)$. Hermite interpolation and the unit T_h imply that T_N is onto for $N \geq d - 1$. The coefficient vector of least norm on $T_N c = u$ is $c_* = T_N^* (T_N T_N^*)^{-1} u$: it has the required image and is orthogonal to $\ker T_N$. Every other vector is $c_* + v$ with $v \in \ker T_N$, so its squared norm is $\|c_*\|^2 + \|v\|^2$. This proves (LR11). In polynomial coordinates the same kernel is exactly $h\mathcal{P}_{N-d}$, including all terms in (LR5). Formula (LR10) gives $M_{N+1} = M_N + tt^*$; multiplication verifies the inverse formula (LR12), and the rank-one determinant identity gives its second formula. No dimension factor was discarded. $\square$

For the tensor-sum source, use its unchanged measure $w_h^{*k} dy$, centre $c = k/2$, and full sum-root polynomial χ_k in the same orthogonal-polynomial construction. In its original polynomial-class coordinates set $t_n = j_{\chi_k}[p_n((S - c)/i)/\sqrt{h_n}]$; if the physical tensor coordinates include a further jet unit, multiply by that exact unit. Hermite interpolation again proves (LR10)–(LR12). This

first gives the attained class metric $G_N = H_N$ in the specified coordinates. For the original observation Λ on this class space, the second minimum is $Q_N = (\Lambda H_N^{-1} \Lambda^*)^{-1}$ on its image. Thus the construction gives entries for the original class metric, rather than confusing it with the larger coefficient Gram before the first minimum. Insert it in $\Omega = \Lambda^* Q_N \Lambda$, $L = H_N - \Omega$ and retain the actual class vectors a_i, v to compute $K_i = a_i^* L v$, $B_i = a_i^* \Omega v$. Their HG7 signed products then keep both cross terms. This is a proved receiving calculation; their growing-degree asymptotics and individual signs have not been evaluated by replacing $q_{S,h} \mathcal{P}_N$ by $\mathcal{P}_N$.

4 Exact source correspondence with the classical zeta heat flow

The original theta source is

$$\phi_*(x) = (4\pi^2 x^4 - 6\pi x^2) e^{-\pi x^2}, \quad f_0(x) = 2 \sum_{n \geq 1} \phi_*(nx), \quad \mathcal{M}f_0 = g.$$

For Polymath's Φ and H_t [4, equations (1)–(4)], direct substitution, with every numerical factor, gives

$$\Phi(u) = \frac{1}{4} e^u f_0(e^{2u}), \quad f_t(x) = e^{t(\log x)^2/4} f_0(x), \quad \mathcal{M}f_t(1/2 + iy) = 16H_t(2y). \quad (\text{LR13})$$

Indeed the two coefficients become $2\pi^2 n^4 e^{9u}$ and $-3\pi n^2 e^{5u}$. On substituting $x = e^{2u}$ in the Mellin integral its Jacobian and $x^{1/2}$ give $8\Phi(u)du$. The evenness of Φ then gives the remaining factor two. The integrals and all derivatives converge by the two-sided superexponential decay supplied by theta inversion. Consequently

$$\partial_t \mathcal{M}f_t(s) = \frac{1}{4} \partial_s^2 \mathcal{M}f_t(s), \quad \partial_t H_t = -\partial_z^2 H_t. \quad (\text{LR14})$$

If B_t denotes multiplication by $e^{t(\log x)^2/4}$ on its domain, direct differentiation gives the exact conductor transformation

$$h(D)B_t = B_t h(D - \frac{t}{2} \log x), \quad [D, \log x] = -I. \quad (\text{LR15})$$

The powers in the right side are operator compositions in their written order; commuting the factors would change the formula. For a simple zero $\rho(t)$ of $g_t = \mathcal{M}f_t$, the implicit function theorem gives its actual velocity

$$\rho'(t) = -\frac{g_t''(\rho(t))}{4g_t'(\rho(t))}. \quad (\text{LR16})$$

Multiple roots continue to require the full primary algebra; (LR16) is asserted only on the simple-root domain. Connes's multiplicative Gaussian in [3] instead acts by $e^{t(s-1/2)^2}$ on the Mellin transform. Its physical convolution and the Gram transport are proved in GM below. Equations (LR13)–(LR15) give the exact relation of the two operations to the same original source and conductor, rather than identifying their different parameter evolutions.

5 Marked Gaussian formula with its arithmetic terms retained

For $t > 0$ and $a \in \mathbb{R}$, let $k_t(v) = e^{-v^2/(4t)}/\sqrt{4\pi t}$ and $F_{t,a}(x) = k_t(\log x - a)$, $f_{t,a} = x^{-1/2} F_{t,a}$. Connes's explicit formula [3, §2, equations (4)–(12)] applies to this rapidly decreasing test function. Its exact marked form is

$$\begin{aligned} \sum_{\rho} m_{\rho} e^{t(\rho-1/2)^2 + a(\rho-1/2)} &= 2e^{t/4} \cosh(a/2) - W_{\mathbb{R}}(F_{t,a}) \\ &\quad - \sum_{n \geq 2} \frac{\Lambda(n)}{\sqrt{n}} [k_t(\log n - a) + k_t(\log n + a)], \end{aligned} \quad (\text{LR17})$$

where the sum on the left ranges over all distinct nontrivial zeros, and Λ is the von Mangoldt function. The full archimedean term is

$$W_{\mathbb{R}}(F) = (\log 4\pi + \gamma_E)F(1) + \int_0^\infty [F(e^u) + F(e^{-u}) - 2e^{-u/2}F(1)] \frac{e^{u/2}}{e^u - e^{-u}} du. \quad (\text{LR18})$$

To prove (LR17), substitute $x = e^v$ and complete the square to obtain $\mathcal{M}f_{t,a}(s) = e^{t(s-1/2)^2 + a(s-1/2)}$. The two pole terms are its values at one and zero. The prime-power terms in Connes's formula are exactly the displayed sum over n , and $x = e^u$ gives (LR18). This proof does not require RH or a self-adjoint zero operator. The zero sum is absolutely and locally uniformly convergent for $t > 0$: $0 < \Re \rho < 1$, Gaussian damping in $\Im \rho$, and the classical $O(T \log(T+2))$ zero count suffice. The prime sum has the same property by Gaussian decay in $\log n$. All fixed-order a derivatives therefore pass through both sums and the integral, by domination with a Gaussian times a polynomial. For example

$$\partial_a^r k_t(u-a) = (2\sqrt{t})^{-r} \mathcal{H}_r((u-a)/(2\sqrt{t})) k_t(u-a), \quad (\text{LR19})$$

where $\mathcal{H}_r$ is the physicists' Hermite polynomial, as follows by differentiating the Gaussian. The derivative of $k_t(u+a)$ gains $(-1)^r$ relative to the same expression. Formula (LR19) keeps the precise growing-order factors. Fixed-order flatness of the prime term as $t \downarrow 0$ does not estimate it after order r grows with the programme degree. For a finite full jet algebra the corresponding marked operator is exactly $j_h \exp(t(s-1/2)^2 + a(s-1/2))$; its nilpotent terms must be kept when applying (LR17) to marked sources.

6 Gaussian heat in the original source, full jets and attained metrics

We use the Gaussian in Alain Connes, *Heat expansion and zeta*, Sections 2–3, original source labels `mellin`, `fourterm`, `ft` and `plicit1` [3]. His multiplicative test function and its Mellin version are, with every factor retained,

$$K_t(v) = \frac{e^{-v^2/(4t)}}{\sqrt{4\pi t}}, \quad F_t(x) = K_t(\log x), \quad f_t(x) = x^{-1/2} F_t(x), \quad t > 0. \quad (\text{GM1})$$

Connes uses $\hat{F}(s) = \int_0^\infty F(x) x^{-is} dx/x$. Consequently $\hat{F}_t(s) = e^{-ts^2}$ and $\mathcal{M}f_t(w) = e^{t(w-1/2)^2}$. The source Hilbert space below has the original measure dx , not dx/x .

6.1 The source operator and the complete physical jet unit

Put $\mathcal{H} = L^2((0, \infty), dx)$, $D = -x\partial_x$, and

$$(UF)(v) = e^{v/2} F(e^v), \quad \mathcal{M}F(w) = \int_0^\infty F(x) x^{w-1} dx.$$

The smooth source space used here is

$$\mathcal{B} = \{F \in C^\infty(0, \infty) : \sup_{x>0} x^a |D^b F(x)| < \infty \text{ for every } a \in \mathbb{R}, b \geq 0\}.$$

Thus UF and each of its derivatives decrease faster than every exponential at both ends of the real line. Keep $g(w) = 2\xi(w) = w(w-1)\pi^{-w/2}\Gamma(w/2)\zeta(w)$ and a full finite divisor $h(w) = \prod_{\rho \in Z} (w-\rho)^{m_\rho}$, where m_ρ is the actual multiplicity of g at ρ . Set $d = \deg h$. The source $a_h \in \mathcal{B}$

is characterized by $\mathcal{M}a_h = g/h$. This characterization preserves the arithmetic unit at every root. Its construction is the following exact division of the original theta source:

$$f_0(x) = 2 \sum_{n \geq 1} (4\pi^2 n^4 x^4 - 6\pi n^2 x^2) e^{-\pi n^2 x^2}, \quad \mathcal{M}f_0 = g, \quad S_\rho F(x) = x^{-\rho} \int_x^\infty F(z) z^{\rho-1} dz.$$

When $\mathcal{M}F(\rho) = 0$, the last integral also equals $-x^{-\rho} \int_0^x F(z) z^{\rho-1} dz$. These two expressions show rapid decrease at the respective endpoints, including every derivative. Differentiating gives $(D - \rho)S_\rho F = F$; integration by parts gives $\mathcal{M}(S_\rho F) = \mathcal{M}F/(w - \rho)$ with its removable value. Applying this operation to the complete multiset of roots produces a_h . Mellin injectivity proves its uniqueness. For completeness, the displayed identity for f_0 follows by integrating its absolutely convergent theta sum for $\Re w > 1$, using $D(D-1)e^{-\pi x^2} = (4\pi^2 x^4 - 6\pi x^2)e^{-\pi x^2}$, and then continuing. Poisson summation for $\sum_{n \in \mathbb{Z}} e^{-\pi n^2 x^2}$ shows that f_0 and every D derivative decrease faster than every power at zero as well as infinity: the two nondecreasing terms x^{-1} and 1 are annihilated by $D(D-1)$. This also verifies $f_0 \in \mathcal{B}$.

Theorem 6 (GM-A: exact Gaussian source and full jet morphism). *The contraction semigroup $T(t)$ on $\mathcal{H}$ defined by $UT(t)U^{-1} = K_t*$ has generator $(D - 1/2)^2$ and satisfies*

$$\begin{aligned} T(t)F(x) &= \int_0^\infty x^{-1/2} z^{-1/2} K_t(\log(x/z)) F(z) dz, \\ \mathcal{M}(T(t)F)(w) &= e^{t(w-1/2)^2} \mathcal{M}F(w) \quad (F \in \mathcal{B}, w \in \mathbb{C}), \\ \|T(t)P(D)a_h\|^2 &= \int_{\mathbb{R}} |P(1/2 + iy)|^2 e^{-2ty^2} w_h(y) dy, \quad w_h(y) = \frac{|(g/h)(1/2 + iy)|^2}{2\pi}. \end{aligned} \tag{GM2}$$

The map $T(t)$ preserves $\mathcal{B}$ and is injective. Let $E_h = \prod_{\rho \in \mathbb{Z}} \mathbb{C}[\tau_\rho]/(\tau_\rho^{m_\rho})$ in the literal coordinates $\tau_\rho = w - \rho$, and let j_h take all divided Taylor coefficients. Define $B_h(t)$ to be multiplication in E_h by $j_h(e^{t(w-1/2)^2})$. Then

$$\begin{aligned} j_h \mathcal{M}(T(t)F) &= B_h(t) j_h \mathcal{M}F, \quad B_h(t) = e^{tL_h}, \quad L_h = j_h((w - 1/2)^2), \\ (B_h(t))_{(\rho,r),(\rho,s)} &= e^{t(\rho-1/2)^2} \sum_{j=0}^{\lfloor (r-s)/2 \rfloor} \frac{(2t(\rho-1/2))^{r-s-2j} t^j}{(r-s-2j)! j!} \quad (r \geq s). \end{aligned} \tag{GM3}$$

Entries with $r < s$ or different roots are zero. In particular this is an invertible full divided-jet map, including all nilpotent terms.

Proof. The substitution $x = e^v$ proves that U is unitary and that $U(D - 1/2)U^{-1} = -\partial_v$. Fourier transformation with the positive exponential e^{iyv} sends this operator to multiplication by iy , and sends K_t* to multiplication by e^{-ty^2} . This proves the generator, contraction, semigroup and injectivity statements, with the closure of the generator understood on its Fourier domain. Substitution $z = e^u$ in the convolution proves its kernel against the original measure dz .

If $|\partial_v^b U F(v)| \leq C_{A,b} e^{-A|v|}$, convolution is bounded by $C_{A,b} e^{-A|v|} \int K_t(u) e^{A|u|} du$; the integral is finite. This applies to every A, b , proves preservation of $\mathcal{B}$ and justifies differentiation and all complex Mellin integrals. Fubini then gives their multiplier $\int K_t(v) e^{(w-1/2)v} dv = e^{t(w-1/2)^2}$. The Gaussian integral is first obtained by completing the square for real $w - 1/2$ and then by entire continuation. Plancherel in the same convention gives $\|F\|^2 = (2\pi)^{-1} \int |\mathcal{M}F(1/2 + iy)|^2 dy$, proving the last identity of (GM2). Finally expand $e^{t(\rho-1/2+\tau)^2} = e^{t(\rho-1/2)^2} e^{2t(\rho-1/2)\tau} e^{t\tau^2}$ modulo τ^{m_ρ} . This proves every entry in (GM3), and the inverse is the same finite algebra exponential at $-t$. $\square$

Fix $N \geq d - 1$. For $u \in E_h$ let $r_h(u)$ be the unique polynomial of degree less than d such that $j_h r_h(u) = j_h(g/h)^{-1}u$. The inverse exists because the multiplicities in h are full. Hermite interpolation here is simply the Chinese remainder map $\mathbb{C}[w]/(h) \simeq E_h$; its kernel consists exactly

of polynomials divisible by every $(w - \rho)^{m_\rho}$, hence by h , and equal finite dimensions give the asserted isomorphism. Put $\mathcal{P}_n = \mathbb{C}[w]_{\leq n}$, with $\mathcal{P}_{-1} = 0$, and keep the coefficient map

$$\begin{aligned} L_N(u, a) &= r_h(u) + ha, & \mathcal{R}_N(u, a) &= L_N(u, a)(D)a_h, & (u, a) &\in E_h \oplus \mathcal{P}_{N-d}, \\ j_h \mathcal{M} \mathcal{R}_N(u, a) &= u, & j_h \mathcal{M} T(t) \mathcal{R}_N(u, a) &= B_h(t)u. \end{aligned} \quad (\text{GM4})$$

Euclidean division by h proves that L_N is an isomorphism. All matrices below use these jet coordinates and the ordinary power coefficients of a . No relation coefficient is discarded.

6.2 All moment derivatives and the moving attained minimum

Let $b_N(y)$ be the row vector specified by $b_N(y)(u, a) = L_N(u, a)(1/2 + iy)$. For $r \geq 0$ set

$$\begin{aligned} M_r(t) &= \int_{\mathbb{R}} y^{2r} e^{-2ty^2} b_N(y)^* b_N(y) w_h(y) dy, & G(t) &= M_0(t), \\ G(t) &= (T(t) \mathcal{R}_N)^* T(t) \mathcal{R}_N = \begin{pmatrix} A(t) & C(t) \\ C(t)^* & S(t) \end{pmatrix}. \end{aligned} \quad (\text{GM5})$$

All these moments are finite: the Mellin transform of $a_h \in \mathcal{B}$ decreases faster than every inverse power on the line in (GM5), by integration by parts in v . Moreover $G(t) > 0$, since the entire nonzero function g/h cannot vanish on an interval and a polynomial vanishing almost everywhere on that line is zero.

Theorem 7 (GM-B: moment derivatives with a signed finite remainder). *For every integer $r \geq 0$ and $t \geq 0$, with right derivatives at zero,*

$$G^{(r)}(t) = (-2)^r M_r(t). \quad (\text{GM6})$$

For every $m \geq 0$,

$$0 \leq (-1)^{m+1} \left[G(t) - \sum_{j=0}^m \frac{(-2t)^j}{j!} M_j(0) \right] \leq \frac{(2t)^{m+1}}{(m+1)!} M_{m+1}(0). \quad (\text{GM7})$$

The exact finite constants

$$\gamma_N = \|G(0)^{-1/2} M_1(0) G(0)^{-1/2}\|, \quad \beta_N = \|G(0)^{-1/2} M_2(0) G(0)^{-1/2}\| \quad (\text{GM8})$$

satisfy, for all $t \geq 0$,

$$e^{-2t\gamma_N} G(0) \preceq G(t) \preceq G(0). \quad (\text{GM9})$$

Proof. Each differentiated entry is bounded by an integrable polynomial moment, proving (GM6) by dominated convergence. For $x \geq 0$ the integral Taylor remainder of e^{-x} after degree m has sign $(-1)^{m+1}$ and absolute value at most $x^{m+1}/(m+1)!$. Multiply this scalar inequality, with $x = 2ty^2$, by $|b_N(y)c|^2 w_h(y)$ and integrate for each coefficient vector c . This proves (GM7) as an inequality of Hermitian forms. For $c \neq 0$, divide this same positive density at $t = 0$ by $c^* G(0) c$ to obtain a probability measure. Its y^2 expectation is at most γ_N . Convexity of $v \mapsto e^{-2tv}$ gives $c^* G(t) c / (c^* G(0) c) \geq e^{-2t\gamma_N}$; the upper bound follows from $e^{-2ty^2} \leq 1$. These are precisely (GM9). $\square$

To include all attained quotients, let J be any fixed surjection from this coefficient space onto a finite coordinate space. Let K be any fixed injective matrix whose image is $\ker J$; empty matrices are allowed when that kernel is zero. Define

$$\begin{aligned} H(t) &= (JG(t)^{-1} J^*)^{-1}, & L(t) &= G(t)^{-1} J^* H(t), & S_K(t) &= K^* G(t) K, \\ u^* H(t) u &= \min_{Jc=u} c^* G(t) c, & JL(t) &= I, & K^* G(t) L(t) &= 0. \end{aligned} \quad (\text{GM10})$$

Indeed the stated lift is feasible, and every other lift is $L(t)u + Ka$. Its norm squared is $u^* H(t) u + a^* S_K(t) a$. This proves the attained minimum and its uniqueness, including every relation variable.

Theorem 8 (GM-C: derivative and Hessian of the full attained form). *The lift and minimum in (GM10) obey*

$$\begin{aligned} L' &= -KS_K^{-1}K^*G'L, \\ H' &= L^*G'L = -2L^*M_1L, \\ H'' &= L^*G''L - 2L^*G'KS_K^{-1}K^*G'L \\ &= 4L^*M_2L - 8L^*M_1KS_K^{-1}K^*M_1L. \end{aligned} \tag{GM11}$$

All matrices on a right-hand side have the same argument t . For the original jet projection $J(u, a) = u$, these formulas are the full Schur identities

$$\begin{aligned} Z &= -S^{-1}C^*, \quad L = \begin{pmatrix} I \\ Z \end{pmatrix}, \quad H = A - CS^{-1}C^*, \\ H' &= A' - C'S^{-1}C^* - CS^{-1}(C')^* + CS^{-1}S'S^{-1}C^*, \\ H'' &= L^*G''L - 2(C' + Z^*S')S^{-1}((C')^* + S'Z). \end{aligned} \tag{GM12}$$

When the relation space is zero the inverse terms are absent. In particular neither cross block nor the motion of its minimizing coefficient is suppressed.

Proof. Differentiate $JL = I$ to see that every column of L' lies in $\text{im } K$. Differentiating $K^*GL = 0$ then gives the first formula in (GM11). Differentiating $H = L^*GL$ makes the terms L'^*GL and L^*GL' vanish, by the same orthogonality. A second derivative gives $L'^*G'L + L^*G''L + L^*G'L'$. Substitution of L' and Hermitian symmetry of G' gives exactly the two correction terms combined in (GM11). Equation (GM6) gives the last expression. The block choice $K = \begin{pmatrix} 0 \\ I \end{pmatrix}$ proves (GM12) by multiplication; alternatively differentiating $Z = -S^{-1}C^*$ gives $Z' = -S^{-1}((C')^* + S'Z)$ and the same Hessian. $\square$

Theorem 9 (GM-D: attained finite error and complete physical jets). *For every quotient (GM10),*

$$\begin{aligned} e^{-2t\gamma_N}H(0) &\preceq H(t) \preceq H(0), \\ -4L(t)^*M_2(t)L(t) &\preceq H''(t) \preceq 4L(t)^*M_2(t)L(t), \\ -R_N(t)H(0) &\preceq H(t) - H(0) - tH'(0) \preceq R_N(t)H(0), \\ R_N(t) &= 4\beta_N \int_0^t (t-s)e^{2s\gamma_N} ds \leq 2\beta_N t^2 e^{2t\gamma_N}. \end{aligned} \tag{GM13}$$

For $\gamma_N > 0$ the exact displayed remainder constant is $R_N(t) = \beta_N(e^{2t\gamma_N} - 1 - 2t\gamma_N)/\gamma_N^2$. For the original projection onto E_h , the metric at a fixed physical smoothed jet v is

$$\begin{aligned} v^*\hat{H}(t)v &= \min_{j_h \mathcal{MT}(t) \mathcal{R}_N c = v} \|T(t)\mathcal{R}_N c\|^2, \\ \hat{H}(t) &= B_h(t)^{-*}H(t)B_h(t)^{-1}, \\ \hat{H}'(t) &= B_h(t)^{-*}[H' - L_h^*H - HL_h]B_h(t)^{-1}, \\ \hat{H}''(t) &= B_h(t)^{-*}[H'' - 2L_h^*H' - 2H'L_h + (L_h^*)^2H + 2L_h^*HL_h + HL_h^2]B_h(t)^{-1}. \end{aligned} \tag{GM14}$$

In particular, with $\ell_N = \|H(0)^{1/2}L_hH(0)^{-1/2}\|$,

$$\begin{aligned} e^{-2t(\gamma_N + \ell_N)}H(0) &\preceq \hat{H}(t) \preceq e^{2t\ell_N}H(0), \\ -\hat{R}_N(t)H(0) &\preceq \hat{H}(t) - H(0) - t\hat{H}'(0) \preceq \hat{R}_N(t)H(0), \\ \hat{R}_N(t) &= 2t^2(\beta_N + 2\ell_N\gamma_N + \ell_N^2)e^{2t(\gamma_N + \ell_N)}. \end{aligned} \tag{GM15}$$

Proof. Take the minimum of (GM9) on the identical affine fibre $Jc = u$. For the Hessian bound, realize the coefficient map as the functions $c \mapsto e^{-ty^2} b_N(y)c$ in $L^2(w_h(y)dy)$. Its restriction to K has Gram S_K . The orthogonal projection onto that finite relation image is therefore this restricted map times S_K^{-1} times its adjoint. Apply the contraction bound for this projection to the function $y^2 e^{-ty^2} b_N(y)L(t)u$. It gives, for every u ,

$$u^* L^* M_1 K S_K^{-1} K^* M_1 L u \leq u^* L^* M_2 L u.$$

Combining it with (GM11) gives the Hessian inequalities in (GM13). Moreover

$$M_2(t) \preceq M_2(0) \preceq \beta_N G(0) \preceq \beta_N e^{2t\gamma_N} G(t), \quad H(t) \preceq H(0).$$

Thus $L(t)^* M_2(t) L(t) \preceq \beta_N e^{2t\gamma_N} H(0)$. Integrating the Hessian bound twice proves the remainder, including its integral constant.

Equation (GM4) says exactly that the feasible coefficient label for physical jet v is $u = B_h(t)^{-1}v$. Substitution into (GM10) proves the first two lines of (GM14). Since $B'_h = L_h B_h = B_h L_h$, ordinary differentiation proves the remaining lines.

To check (GM15), perform all estimates in the fixed norm $\|u\|_{H(0)}^2 = u^* H(0) u$. The power series and submultiplicativity give $\|e^{\pm t L_h}\|_{H(0)} \leq e^{t\ell_N}$. In particular $\|e^{-t L_h} u\|_{H(0)} \geq e^{-t\ell_N} \|u\|_{H(0)}$ by applying the bound for the inverse. This and (GM13) prove the first line. The same comparison with M_1 in place of M_2 gives $\|H(0)^{-1/2} H'(t) H(0)^{-1/2}\| \leq 2\gamma_N e^{2t\gamma_N}$. The Hessian estimate already gives the analogous bound $4\beta_N e^{2t\gamma_N}$ for H'' ; also $\|H(0)^{-1/2} H(t) H(0)^{-1/2}\| \leq 1$. Consequently the bracket in the last line of (GM14) has norm at most $4(\beta_N + 2\ell_N \gamma_N + \ell_N^2) e^{2t\gamma_N}$ in these fixed coordinates. Congruence by B_h^{-1} adds at most $e^{2t\ell_N}$. Integrating twice and bounding the exponential by its endpoint proves the stated $\hat{R}_N$. $\square$

These estimates also cover any specified finite sequence of fixed restrictions and attained quotients. Here is the exact reason. A fixed restriction replaces the source coefficient space by a fixed linear subspace. A quotient minimizes over a fixed affine fibre in that subspace. Restricting its target to a further fixed subspace restricts the preceding source to its inverse image; taking a further quotient composes the fixed surjections. Thus, by induction, each resulting metric is a single minimum of the original form on a fixed subspace with a fixed surjection. Apply the proofs above to that subspace. The inequalities $M_j(0) \preceq c_j G(0)$ restrict with the same constants, so (GM13) still holds with γ_N, β_N from the full original source. Metric-dependent changes of frame are not used in this statement.

For a rank- r result $Q(t)$ in such fixed coordinates, (GM13) implies

$$-2rt\gamma_N \leq \log \frac{\det Q(t)}{\det Q(0)} \leq 0. \quad (\text{GM16})$$

Indeed conjugation by $Q(0)^{-1/2}$ puts all its eigenvalues in $[e^{-2t\gamma_N}, 1]$, and their logarithms add. Rank zero has determinant 1. Therefore a finite specified return $\mathcal{L}(t) = \sum_\nu a_\nu \log \det Q_\nu(t)$ obeys $|\mathcal{L}(t) - \mathcal{L}(0)| \leq 2t \sum_\nu |a_\nu| r_\nu \gamma_{N_\nu}$. All endpoint coefficients and ranks remain in this error. For physical jets the exact additional determinant term is

$$\log \frac{\det \hat{H}(t)}{\det H(0)} = \log \frac{\det H(t)}{\det H(0)} - 2t \Re \sum_{\rho \in Z} m_\rho (\rho - 1/2)^2. \quad (\text{GM17})$$

This follows because L_h is triangular with diagonal entry $(\rho - 1/2)^2$ repeated m_ρ times. It is an exact unit contribution, not an omitted scalar normalization.

6.3 Tensor heat, sum heat and the original complete quartet

Retain k separate physical variables, $D_j = -x_j \partial_{x_j}$, and $\mathcal{H}^{\otimes k} = L^2((0, \infty)^k, dx_1 \cdots dx_k)$. Write $A_j = D_j - 1/2$ and $S = \sum_j w_j$, with $w_j = 1/2 + iy_j$. Define the three commuting contraction semigroups by their Fourier multipliers:

	generator	multiplier at $(y_1, \dots, y_k)$	
$T_{\text{fac}}(t)$	$\sum_j A_j^2$	$e^{-t \sum_j y_j^2}$	(GM18)
$T_{\text{sum}}(t)$	$(\sum_j A_j)^2$	$e^{-t(\sum_j y_j)^2}$	
$T_{\perp}(t)$	$\sum_j A_j^2 - k^{-1}(\sum_j A_j)^2$	$e^{-t \sum_j (y_j - k^{-1} \sum_i y_i)^2}$	

In particular $T_{\text{fac}}(t) = T(t)^{\otimes k}$. The original additive source is

$$\mathcal{A}_{k,N} P = P(D_1 + \cdots + D_k)(a_h \otimes \cdots \otimes a_h), \quad P \in \mathbb{C}[S]_{\leq N}.$$

Its multiple Mellin transform is exactly $P(S) \prod_j (g/h)(w_j)$; this follows by expanding each monomial with its original multinomial coefficients and using integration by parts in every variable.

Theorem 10 (GM-E: exact tensor-to-sum heat map and moment receiver). *There is the bounded operator identity*

$$T_{\text{fac}}(t) = T_{\text{sum}}(t/k) T_{\perp}(t). \quad (\text{GM19})$$

The complete Mellin multipliers of the smoothed additive sources are respectively

$$e^{t \sum_j (w_j - 1/2)^2} P(S) \prod_j (g/h)(w_j), \quad e^{t(S - k/2)^2} P(S) \prod_j (g/h)(w_j). \quad (\text{GM20})$$

Their original power-coefficient Gram forms are

$$\begin{aligned} P^* \mathcal{G}_{\text{fac}}(t) P &= \int_{\mathbb{R}} |P(k/2 + iy)|^2 (e^{-2t(\cdot)^2} w_h)^{*k}(y) dy, \\ P^* \mathcal{G}_{\text{sum}}(t) P &= \int_{\mathbb{R}} |P(k/2 + iy)|^2 e^{-2ty^2} w_h^{*k}(y) dy, \\ \mathcal{G}_{\text{fac}}(t) &\preceq \mathcal{G}_{\text{sum}}(t/k). \end{aligned} \quad (\text{GM21})$$

All of (GM6)–(GM13) apply to each of these two full Grams and any of its fixed attained quotients, with their respective original moments

$$M_r^{\diamond}(t) = \int_{\mathbb{R}^k} \eta_{\diamond}(\mathbf{y})^r e^{-2t\eta_{\diamond}(\mathbf{y})} b_{k,N}(\mathbf{y})^* b_{k,N}(\mathbf{y}) \prod_j w_h(y_j) d\mathbf{y}, \quad \eta_{\text{fac}} = \sum_j y_j^2, \quad \eta_{\text{sum}} = \left(\sum_j y_j\right)^2, \quad (\text{GM22})$$

where $b_{k,N}(\mathbf{y})$ is the row evaluating the original S -power basis at $S = k/2 + i \sum_j y_j$.

Proof. The identity $\sum_j y_j^2 = k^{-1}(\sum_j y_j)^2 + \sum_j (y_j - k^{-1} \sum_i y_i)^2$ proves (GM19) pointwise on the unitary multiple Fourier transform. Each factor is bounded, so this proves an equality on the whole Hilbert space, with no inverse heat operator required. The entire Mellin version gives (GM20), first on tensor products of $\mathcal{B}$ and their finite derivative sums, which include the displayed sources. The same Gaussian convolution argument as in GM-A justifies this at all complex coordinates. Multiple Plancherel gives the k -fold integrals of $|P(k/2 + i \sum_j y_j)|^2$ against the two exponential densities. Integrate with $y = \sum_j y_j$ and $y_k = y - \sum_{j < k} y_j$. This change of variables has determinant one, and Tonelli gives the two convolutions in (GM21). The remaining factor $e^{-2t \sum_j (y_j - y/k)^2} \leq 1$ proves their comparison. Every polynomial moment in (GM22) is finite, since it is bounded by a finite sum of products of the one-factor moments. Positivity follows from the same polynomial argument as for (GM5). The proofs of GM-B–D use only a nonnegative measurable multiplier η , its moments, and projection onto the actual relation space. Replacing y^2 by the explicitly displayed $\eta_{\diamond}$ proves each formula, without changing its factors of two or its minima. $\square$

For example the full tensor coefficient space $\mathcal{P}_N^{\otimes k}$ gives valid bounds $\gamma_{\text{fac}} \leq k\gamma_N$ and $\gamma_{\text{sum}} \leq k^2\gamma_N$ after its exact restriction to the additive source. Here γ_N can be computed in either original coefficient frame, since its definition is the supremum of the same Rayleigh quotient. Indeed each insertion of the one-factor first moment is at most γ_N times the full tensor Gram, and $(\sum_j y_j)^2 \leq k \sum_j y_j^2$. Likewise $\beta_{\text{fac}} \leq k^2\beta_N$ and $\beta_{\text{sum}} \leq k^4\beta_N$, using $(\sum_j y_j^2)^2 \leq k \sum_j y_j^4$. These inequalities explicitly retain both k and the degree-dependent original constants.

Now specialize to the original quartet, writing its positive displacement as δ_ρ throughout:

$$\begin{aligned} \rho &= \frac{1}{2} + \delta_\rho + i\gamma, \quad 0 < \delta_\rho < \frac{1}{2}, \quad \gamma > 2, \quad m_0 \geq 1, \quad k \equiv 1 \pmod{4}, \quad k \geq 5, \\ h(w) &= \prod_{\epsilon, \eta \in \{1, -1\}} [w - \frac{1}{2} - \epsilon\delta_\rho - i\eta\gamma]^{m_0}, \quad e = 1 + k(m_0 - 1), \quad q = e(k + 1)^2, \\ \kappa_{a,b} &= k/2 + (2a - k)\delta_\rho + i(2b - k)\gamma, \quad \chi(S) = \prod_{a,b=0}^k (S - \kappa_{a,b})^e. \end{aligned} \tag{GM23}$$

This calculation studies the stipulated off-axis quartet with its given parameters; it does not assert that zeta has such a quartet. Let $\mathcal{T}_h = E_h^{\otimes k}$ and let $s = \sum_j w_j$ in this finite algebra. Multiplication by the full arithmetic unit $\prod_j j_h(g/h)(w_j)$ is denoted $\mathcal{U}_h$.

Theorem 11 (GM-F: complete sum embedding and exact descent exceptions). *There is an injective algebra map*

$$\iota : E_\chi = \mathbb{C}[S]/(\chi) \longrightarrow \mathcal{T}_h, \quad [P] \longmapsto P(s). \tag{GM24}$$

The physical full jet maps of the three additive sources (unheated, factor heat, sum heat) are exactly

$$\begin{aligned} &\mathcal{U}_h \iota([P]), \quad \mathcal{U}_h Q_t \iota([P]), \quad \mathcal{U}_h \iota(E_t[P]), \\ Q_t &= \exp \left(t \sum_j (w_j - 1/2)^2 \right), \quad E_t = \exp(t(S - k/2)^2) \in E_\chi. \end{aligned} \tag{GM25}$$

Finite algebra exponentials here retain all nilpotent coefficients. For $t > 0$, the factor unit Q_t belongs to $\iota(E_\chi)$ exactly in the following case:

$$m_0 = 1 \quad \text{and} \quad 4t\delta_\rho\gamma \in \pi\mathbb{Z}. \tag{GM26}$$

For $m_0 \geq 2$ it never belongs to that subalgebra at positive time. For every m_0, t the multiplication map

$$\mathcal{U}_h \iota(E_\chi) \longrightarrow \mathcal{U}_h Q_t \iota(E_\chi), \quad v \longmapsto Q_t v \tag{GM27}$$

is an explicitly invertible linear map, with inverse multiplication by Q_{-t} . Thus failure of descent does not remove the exact correspondence between the original physical jet images.

Proof. Decompose $\mathcal{T}_h$ into its primary factors, one for each ordered tuple of quartet roots. At such a tuple write $w_j = 1/2 + \epsilon_j\delta_\rho + i\eta_j\gamma + \tau_j$, with $\tau_j^{m_0} = 0$. The factor is $\mathbb{C}[\tau_1, \dots, \tau_k]/(\tau_1^{m_0}, \dots, \tau_k^{m_0})$. If precisely a of the ϵ_j and b of the η_j equal 1, then $s = \kappa_{a,b} + \sum_j \tau_j$. The latter nilpotent sum has index exactly e : its e -th power is zero by degree, while

$$(\sum_j \tau_j)^{e-1} = \frac{(k(m_0-1))!}{((m_0-1)!)^k} \prod_j \tau_j^{m_0-1} \neq 0.$$

Every pair (a, b) occurs, by choosing independently the sets of positive ϵ_j and positive η_j . The $\kappa_{a,b}$ are distinct, since their real and imaginary differences vanish only when both indices agree. The minimal polynomial of s on the whole product is therefore exactly χ , proving (GM24) including every primary multiplicity. Taking divided Taylor coefficients of (GM20) proves (GM25); the factor $\mathcal{U}_h$ is a unit because every factor $j_h(g/h)$ is a unit.

First take $m_0 = 1$. A function on the ordered root tuples is in $\iota(E_\chi)$ precisely when it is constant on tuples with the same pair (a, b) : necessity follows from evaluation at s , and

sufficiency follows by polynomial interpolation at the distinct $\kappa_{a,b}$. If c is the number of indices with $\epsilon_j = \eta_j = 1$, direct counting gives

$$\sum_j (\epsilon_j \delta_\rho + i \eta_j \gamma)^2 = k(\delta_\rho^2 - \gamma^2) + 2i \delta_\rho \gamma (k - 2a - 2b + 4c).$$

Thus (GM26) makes the exponential independent of c and proves sufficiency. For necessity, compare two tuples with the same last $k-2$ roots, but first two sign pairs respectively $(++)$, $(--)$ and $(+-)$, $(-+)$. Their sums are identical and the ratio of their Q_t values is $e^{8it\delta_\rho\gamma}$. Constancy forces this ratio to be one, which is exactly (GM26). At these resonant times the descended unit has at $\kappa_{a,b}$ the nonzero value

$$\exp\{tk(\delta_\rho^2 - \gamma^2) + 2it\delta_\rho\gamma(k - 2a - 2b)\}.$$

Its unique interpolation polynomial of degree below q specifies it completely; it need not equal the different unit E_t .

If $m_0 \geq 2$, choose a tuple with two distinct roots, say the first two. Every element $p(s)$ has identical coefficients of τ_1 and τ_2 , both $p'(\kappa_{a,b})$. The respective coefficients of Q_t are its common nonzero constant coefficient multiplied by $2t(\epsilon_1\delta_\rho + i\eta_1\gamma)$ and by $2t(\epsilon_2\delta_\rho + i\eta_2\gamma)$. They are unequal for $t > 0$, since these roots are distinct. Those first coefficients survive modulo $\tau_j^{m_0}$, proving nonmembership, even at the resonant times of the simple-root case. Finally $Q_t Q_{-t} = 1$ in the finite algebra, and it commutes with U_h . This proves (GM27) with its stated inverse. $\square$

For $N \geq q-1$, keep the original remainder map $J_{\chi,N} : \mathbb{C}[S]_{\leq N} \rightarrow E_\chi$ in its power coordinates. Its kernel is the complete ideal $\chi\mathbb{C}[S]_{\leq N-q}$. Equation (GM25) shows that this kernel is unchanged under either Gaussian heat: both physical maps are the original injective sum embedding followed by units. Hence the *labelled* attained forms are exactly

$$H_\diamond(t) = (J_{\chi,N} \mathcal{G}_\diamond(t)^{-1} J_{\chi,N}^*)^{-1} \quad (\diamond = \text{fac, sum}), \quad (\text{GM28})$$

and GM-C-D compute their derivatives and remainders over every coefficient of that ideal. For sum heat, the metric at a fixed physical vector $U_h \iota(v)$ is $E_t^{-*} H_{\text{sum}}(t) E_t^{-1}$ in the original v coordinates; GM14–15 apply with L_h replaced by multiplication by $(S - k/2)^2$ in the full algebra E_χ . For factor heat, the physical vector with label v is instead $U_h Q_t \iota(v)$ and belongs to the moving subspace in (GM27). Outside the exact cases (GM26) a congruence by a putative sum-coordinate unit is false. The full tensor map (GM27) is its correct replacement, and (GM28) remains the attained metric of every such moving labelled vector.

6.4 The precise relation to Connes's scalar Gaussian family

For original coefficient vectors c, d define the linear functional on bounded functions of y by

$$\mathfrak{L}_{c,d}(\varphi) = \int_{\mathbb{R}} \varphi(y) \overline{b_N(y)} c b_N(y) d w_h(y) dy.$$

It is well-defined by Cauchy–Schwarz and (GM5), and the exact Gaussian receiver is

$$c^* G(t) d = \mathfrak{L}_{c,d}(\widehat{F}_{2t}), \quad \widehat{F}_{2t}(y) = e^{-2ty^2}. \quad (\text{GM29})$$

The factor $2t$ is forced by the norm square. Connes's zero-side functional, instead, evaluates $\mathcal{M}f_t$ at zeta zeros: its Gaussian summand at a zero ρ is $\mathcal{M}f_t(\rho) = e^{t(\rho-1/2)^2}$, with the original zero multiplicity. On a finite set of zeros this follows immediately from (GM1); the full explicit-formula extension is Connes's Mellin formula in Section 2 [3], with its archimedean and prime terms retained. His interpretation as a heat trace of a self-adjoint operator with zeta ordinates

assumes RH, as stated in his Introduction and Theorem 1.1. The original-source identities (GM1)–(GM29) make no such assumption: they pair the same explicit Gaussian transform with the explicitly different source functional.

The source heat $T(t)$ is not trace class for $t > 0$. In its unitary Fourier realization choose disjoint measurable subsets of $[-1, 1]$ of positive measure and their normalized indicators. They form an orthonormal sequence. Their images under multiplication by e^{-ty^2} have pairwise disjoint supports and norms at least e^{-t} . Thus the images have no convergent subsequence, proving that $T(t)$ is not compact; every trace-class operator is compact. Consequently the source Gram in (GM29) is not a trace of this heat operator and is not obtained by declaring a unitary identification with Connes's discrete trace. The exact transforms (GM2), (GM20) and (GM29), and the finite jet maps (GM3), (GM25)–(GM28), specify the actual relations used here.

For each specified finite N, k and divisor, the constants in (GM8), (GM15) and (GM22) are finite and computed from the full original integrals and units. For a finite determinant return, (GM16) gives an attained error tolerance by choosing $t \leq \varepsilon / (2 \sum_\nu |a_\nu| r_\nu \gamma_{N_\nu})$ when that denominator is positive; a zero denominator gives zero error. This is a finite calculation and convergence statement. No bound uniform in a growing degree, tensor order, packet size or moving divisor is asserted. In particular a fixed-order scalar heat asymptotic from Connes is not substituted for growth estimates on these actual moment matrices or their attained current receivers.

7 The observed terminal class on the complete original window

This section receives the new continuation's equations (1)–(34). It verifies its minimum argument in the original coordinates and strengthens its arithmetic–Gamma same-class comparison. The period nonvanishing calculation is proved separately in OC below.

Retain a simple hypothetical zeta quartet with $\rho = 1/2 + \delta + i\gamma$, $0 < \delta < 1/2$, and the original fixed period ϖ . Let $k \equiv 1 \pmod{4}$, $k \geq 9$, $q = (k+1)^2$, $c = k/2$, and

$$\chi_k(S) = \prod_{a,b=0}^k [S - c - (2a - k)\delta - i(2b - k)\gamma], \quad v_+(S) = \frac{\chi_k(S)}{(S - k\rho)\chi'_k(k\rho)}. \quad (\text{OB1})$$

The roots are distinct. The physical source metric is $dm_k(y) = w_h^{*k}(y)dy$, $S = c + iy$, with w_h in (LR2). Let $E_k = \mathbb{C}[S]/(\chi_k)$ and G_N its attained metric at polynomial degree at most N . The observation $\Lambda_k : E_k \rightarrow B_k$ is the original period-dependent invariant observation, onto its image. It includes the original conductor-multiple invariant subspace, as expressed by FW46. Define

$$E_N = v_+^* G_N v_+, \quad Q_N = (\Lambda_k G_N^{-1} \Lambda_k^*)^{-1}, \quad e_{B,N} = (\Lambda_k v_+)^* Q_N \Lambda_k v_+. \quad (\text{OB2})$$

Both minima include every polynomial relation and kernel lift. In particular, $e_{B,N}$ is the minimum of $\|P\|_{m_k}^2$ over $\deg P \leq N$, $\Lambda_k[P] = \Lambda_k v_+$, and $e_{B,N} = E_N(1 - \theta_N)$ is the exact FW38 angle identity.

Set $j = k - 8$, $q_- = (k - 7)^2$, $\Delta = q - q_- = 16k - 48$ and $c_- = c - 4$. The lower cardinal v_- uses χ_j and $j\rho$ in (OB1). The conductor $\mathcal{T}_A$ has the literal coefficients (AC1), without multiplication by $v_A!/\mu_{v_A}$, where $v_A = \text{ord}_0 E_A \leq 80$. The exact maps are

$$C[P] = [\mathcal{T}_A P] \in E_j, \quad \ker \Lambda_k \subset \ker C, \quad C v_+ = a_{88} v_-. \quad (\text{OB3})$$

The first assertion descends because a lower root plus any degree-eight shift is an upper root. The inclusion is the transpose of inclusion of every conductor-multiple lower biform into the original invariant observation image. For the last assertion, evaluation at every lower root gives zero except at $j\rho$, where only the $(8, 8)$ shift reaches $k\rho$. This proves its value, including its coefficient. The degree drops by at least v_A by the exponential Taylor expansion.

7.1 A consecutive-minimum estimate retaining the entire residual

Use the original reference $d\sigma = |\Gamma(1/4 + iy/2)|^2 dy/(2\pi)$. Put $Q_-(y) = i^{-q_-} \chi_j(c_- + iy)$, $dW_- = Q_-^2 d\sigma$. The polynomial Q_- is real and even; since j is odd, every root has imaginary part of modulus at least δ . It has no real zero. Let U_r be the real monic orthogonal polynomials for W_- , $\nu_r = \|U_r\|_{W_-}^2$, and $a_r = \sqrt{\nu_r/\nu_{r-1}}$. Their recurrence is $yU_r = U_{r+1} + a_r^2 U_{r-1}$.

Lemma 12 (Two bounds for every recurrence coefficient). *For $r \geq 1$, with $B_- = 2q_-/\delta + 2 + \pi/2$,*

$$r/B_- \leq a_r \leq 2(q_- + r). \quad (\text{OB4})$$

Proof. The digamma series [6, 5.7.6] gives $\Im\psi(1/4 + ib) = \sum_{n \geq 0} b/((n + 1/4)^2 + b^2)$. The first term has modulus at most two. For $|b| > 0$, the remaining positive decreasing terms are bounded by $\int_{1/4}^\infty |b|/(u^2 + b^2) du \leq \pi/2$; at zero they vanish. Differentiating the actual Gamma factor gives $(\log \sigma)'(y) = -\Im\psi(1/4 + iy/2)$, so its modulus is at most $2 + \pi/2$. Also $|Q'_-/Q_-| \leq q_-/\delta$ on the whole real line. Thus $|W'_-/W_-| \leq B_-$. Integration of $(U_r U_{r-1} W_-)'$ has zero boundary values by exponential Gamma decay. Orthogonality gives $r\nu_{r-1} = -\int U_r U_{r-1} W'_-$; Cauchy–Schwarz proves the lower bound. For the upper bound, yU_{r-1} is a monic trial. The exact Gamma recurrence has adjacent entries $\sqrt{(n+1)(n+1/2)}$ and $\sqrt{n(n-1/2)}$, hence multiplication on degree n has norm at most $2(n+1)$, including its output degree. Apply it to $Q_- U_{r-1}$, whose degree is $q_- + r - 1$. $\square$

Let $z_- = j\gamma - ij\delta$, $d_- = j\delta$, $R_- = |z_-|$ and

$$H_r = \min_{\deg P < r} \int \left| \frac{1}{z_- - y} - P(y) \right|^2 dW_-(y), \quad r \geq 0. \quad (\text{OB5})$$

The empty minimum at $r = 0$ means no polynomial is subtracted.

Lemma 13 (Consecutive full residual).

$$\frac{a_{r+1}^2}{R_-^2 + a_{r+1}^2 + a_{r+2}^2} \leq H_{r+1}/H_r \leq \frac{a_{r+1}^2}{d_-^2 + a_{r+1}^2}. \quad (\text{OB6})$$

Proof. The measure W_- has an exponential moment near zero. Polynomials are complete: an orthogonal vector times W_- has an analytic bilateral transform in a strip by Cauchy–Schwarz; all derivatives at zero vanish; the identity theorem and Fourier uniqueness give zero. In the orthonormal polynomial basis, multiplication by y is a self-adjoint Jacobi operator J . The coefficients of $(z_- - y)^{-1}$ are $\sqrt{\nu_0} w$, $w = (z_- - J)^{-1} e_0$. Therefore $H_r = \nu_0 \sum_{n \geq r} |w_n|^2$.

Remove the single edge of size a_{r+1} between indices $r, r+1$. This bounded self-adjoint finite-rank perturbation splits off a finite prefix and a self-adjoint tail $J^{(r+1)}$. Its resolvent equation is $w_{>r} = a_{r+1} w_r (z_- - J^{(r+1)})^{-1} e_0$. The coefficient w_r cannot vanish: then this tail vanishes, and successive recurrence equations force $w_0 = 0$ and contradict the zeroth equation $(z_- - J)w = e_0$. If $L = \|(z_- - J^{(r+1)})^{-1} e_0\|^2$, it follows exactly that $H_{r+1}/H_r = a_{r+1}^2 L / (1 + a_{r+1}^2 L)$. The spectral probability of its unit vector e_0 has first moment zero and second moment a_{r+2}^2 . Jensen applied to the positive variable $|z_- - y|^2$ gives $L \geq (R_-^2 + a_{r+2}^2)^{-1}$; the self-adjoint resolvent estimate gives $L \leq d_-^2$. Monotonicity of $x/(1+x)$ proves (OB6). The original mass ν_0 has cancelled between the two norms, not been replaced by one. $\square$

Write $\mathcal{E}_-^\sigma(L)$ for the attained norm of v_- at degree L and centre c_- . Since $\chi'_j(j\rho) = i^{q_- - 1} Q'_-(z_-)$,

$$v_-(c_- + iy) = \frac{Q_-(y)}{(y - z_-) Q'_-(z_-)}, \quad \mathcal{E}_-^\sigma(q_- - 1 + r) = H_r / |Q'_-(z_-)|^2. \quad (\text{OB7})$$

The free relation polynomials have degree below r , which verifies this is the whole affine minimum. For $q - 1 \leq N \leq 2q$, put

$$g_{k,N} = \{1 + B_-^2 [R_-^2 + 4(N - v_A + 3)^2]\}^{-1}. \quad (\text{OB8})$$

In the $\Delta - v_A$ steps between $N - \Delta$ and $N - v_A$, (OB4) and (OB6) bound each ratio below by (OB8): use $r + 1 \geq 1$ and $q_- + r + 2 \leq N - v_A + 3$. The initial degree is at least $q_- - 1$. Therefore

$$\mathcal{E}_-^\sigma(N - v_A) \geq g_{k,N}^{\Delta - v_A} \mathcal{E}_-^\sigma(N - \Delta). \quad (\text{OB9})$$

7.2 Exact reverse lift and finite observed lower bound

Let $\lambda = k\rho$ and $D_k(S) = \chi_k(S)/\chi_j(S - 8\rho)$. This is a monic polynomial: translation identifies the lower grid with the upper subgrid $8 \leq a, b \leq k$. Its degree is Δ and

$$v_+(S) = \frac{D_k(S)}{D_k(\lambda)} v_-(S - 8\rho), \quad D_k(\lambda) = \prod_{\substack{0 \leq r, s \leq k \\ (r, s) \notin [0, k-8]^2}} (2r\delta + 2is\gamma). \quad (\text{OB10})$$

These follow by differentiating the exact polynomial factorization at λ . Every factor in the product has modulus at least $2(k - 7)\delta$. For each lower representative P_- of degree $N - \Delta$, $D_k(S)P_-(S - 8\rho)/D_k(\lambda)$ is an upper representative of degree N , including all of its relation terms.

For a Gamma polynomial of degree at most n one has the explicit bound

$$\|P(\cdot + z)\|_\sigma \leq \mathcal{T}_n(z) \|P\|_\sigma, \quad \mathcal{T}_n(z) = e(n+1)(2n+1)^{(|z|+1)/2}. \quad (\text{OB11})$$

Here is a finite proof. Its generating function is $(1 + t^2)^{-1/4} e^{y \arctan t}$, by the Gamma recurrence. Translation multiplies it by $e^{z \arctan t}$. On $|t| = r < 1$, $|\arctan t| \leq \frac{1}{2} \log((1+r)/(1-r))$. For $n \geq 1$, use $r = n/(n+1)$ in Cauchy's coefficient formula to bound each of the first $n+1$ multiplier coefficients by $e(2n+1)^{|z|/2}$. The weighted upper diagonal at gap l has norm at most $\sqrt{2l+1}$, from $\rho_{i+l}/\rho_i \leq \sqrt{2l+1}$ with $\rho_n^2 = n!/(1/2)_n$. Summing the $n+1$ diagonal norms gives (OB11). At $n = 0$ translation fixes the constant and $\mathcal{T}_0 = e$. The mass $\sqrt{2\pi}$ appears on both sides of the same norm.

The displacement of the reverse lift is $z_0 = -8\gamma + 8i\delta$. Each root of $D_k(c + iy)$ has modulus at most $R = k\sqrt{\delta^2 + \gamma^2}$ in the y coordinate. Successive Gamma multiplication bounds, with all degrees at most N , prove

$$\mathcal{E}_+^\sigma(N) \leq \frac{\mathcal{T}_{N-\Delta}(z_0)^2 [2(N+1) + R]^{2\Delta}}{|D_k(\lambda)|^2} \mathcal{E}_-^\sigma(N - \Delta). \quad (\text{OB12})$$

Use CAI16–18 at degree $2q+2$, setting $\ell_k = a_k/[1+4(2q+2+M)^2]^M$, $u_k = A_k$, $M = 25$ on this simple-root domain. They are the original positive constants and give $\ell_k \|P\|_\sigma^2 \leq \|P\|_{m_k}^2 \leq u_k \|P\|_\sigma^2$ uniformly on the window; $\log(u_k/\ell_k) = O_h(k + \log q)$. In the two original centres the conductor bound is

$$\|\mathcal{T}_A P\|_{\sigma, c_-} \leq \mathfrak{C}_{A,N} \|P\|_{\sigma, c}, \quad \mathfrak{C}_{A,N} = \sum_{u,w=0}^8 |a_{uw}| \mathcal{T}_N((2w-8)\gamma - i(2u-8)\delta). \quad (\text{OB13})$$

This follows by applying (OB11) to each original summand. Every observation lift in (OB2) must have conductor class $a_{88}v_-$ by (OB3). Its Gamma norm is therefore at least $|a_{88}|^2 \mathcal{E}_-^\sigma(N - v_A)/\mathfrak{C}_{A,N}^2$. Take the original full minimum and combine with (OB9)–(OB12) to obtain

$$1 - \theta_N \geq \mathfrak{l}_{k,N} := \frac{\ell_k}{u_k} \frac{|a_{88}|^2 |D_k(\lambda)|^2 g_{k,N}^{\Delta - v_A}}{\mathfrak{C}_{A,N}^2 \mathcal{T}_{N-\Delta}(z_0)^2 [2(N+1) + R]^{2\Delta}} > 0. \quad (\text{OB14})$$

The positivity holds on the original large-period domain proved in OC; no corner coefficient is assigned by hand. On that domain all constants depending on h, ϖ are fixed before k grows. Since $\Delta = O(k)$, $-\log g_{k,N} = O_h(\log q)$ and the translations have fixed exponents, (OB14) proves

$$0 \leq -\log(1 - \theta_N) \leq -\log \mathfrak{l}_{k,N} = O_{h,\varpi}(k \log q), \quad q - 1 \leq N \leq 2q. \quad (\text{OB15})$$

In particular $\mathfrak{l}_{k,N} \leq 1$ follows from its proved comparison with the actual energy ratio; it need not be separately imposed.

7.3 Receiving the estimate and strengthening arithmetic comparison

Insert (OB15) in the proved full-source FW35–36. With $B_N = -\log \mathfrak{l}_{k,N}$, the four-angle logarithm lies in $[-B_{q-1} - B_q, B_{2q-1} + B_{2q}]$. Thus

$$\log \frac{e_{B,q-1} e_{B,q}}{e_{B,2q-1} e_{B,2q}} = C_{\partial} q + O_{h,\varpi}(k \log q). \quad (\text{OB16})$$

Each single window has half the leading coefficient. Since $k \log q = o(q/\log q)$, the angle contributes at neither displayed scale. For fixed $0 < s \leq 1$, the same substitution in FW43 gives

$$-\log \frac{e_{B,q-1+\lfloor sq \rfloor}}{e_{B,q-1}} = q\mathcal{J}(s) + O_{h,\varpi,s}(k \log q), \quad \mathcal{J}(s) = \int_0^s f(t) dt. \quad (\text{OB17})$$

Here the unchanged explicit FW44 profile is $f(t) = \log((1+r_t)/(1-r_t))$, $r_t = \sqrt{1-\kappa_t^2}$, $E(\kappa_t)/(r_t K(\kappa_t)) = 1+t$; its logarithmic singularity at zero is integrable, and $\mathcal{J}(0) = 0$. Here the complete elliptic integrals retain their usual modulus convention

$$K(\kappa) = \int_0^{\pi/2} \frac{d\alpha}{\sqrt{1-\kappa^2 \sin^2 \alpha}}, \quad E(\kappa) = \int_0^{\pi/2} \sqrt{1-\kappa^2 \sin^2 \alpha} d\alpha.$$

The original endpoint solution has $0 < \kappa_t < 1$ for $t > 0$. Equivalently the fully specified constant in (OB16) is $C_{\partial} = 2\mathcal{J}(1) = 2 \int_0^1 f(t) dt$, by the two single-window identities in FW35 and FW43. This is the same original boundary constant, with no change of modulus or source measure.

There is a sharper arithmetic comparison than the intake's equation (31), valid for every fixed nonzero observed class, without a corner restriction. Minimize the two whole-polynomial CAI bounds on exactly the same observation fibre. At each original cutoff this gives

$$\ell_k e_{B,N}^{\sigma} \leq e_{B,N}^{\text{ar}} \leq u_k e_{B,N}^{\sigma}. \quad (\text{OB18})$$

For the four-cutoff logarithmic returns $\mathcal{E}_B^{\text{ar}}$ and $\mathcal{E}_B^{\sigma}$, sum the two numerator inequalities and subtract the two denominator inequalities to get

$$|\mathcal{E}_B^{\text{ar}} - \mathcal{E}_B^{\sigma}| \leq 2 \log(u_k/\ell_k) = O_h(k + \log q). \quad (\text{OB19})$$

The common source mass cancels only in this equal-sign-count expression. For a single prefix ratio the bound is $\log(u_k/\ell_k)$. This strengthens the incoming $O(k \log q)$ comparison and is uniform over all nonzero observation classes, even where (OB14) is not invoked.

Finally set $b_r^B = 1 - e_{B,N+1}/e_{B,N}$, $r = N+1-q$. The original rank-one update FW40 identifies it, on its defined quotient domain, as $T_r|1-V_N|^2/[d_{K,N}(1-\theta_N)]$; the undivided energy formula defines it also when a displayed response denominator vanishes. Positive nested minima give $0 \leq b_r^B < 1$. Telescoping proves

$$-\sum_{r=0}^q w_r \log(1-b_r^B) = C_{\partial} q + O_{h,\varpi}(k \log q), \quad w_0 = w_q = 1, \quad w_r = 2 \ (0 < r < q). \quad (\text{OB20})$$

Moreover $q^{-1} \sum_{r=0}^{q-1} -\log(1-b_r^B) \delta_{r/q}$ converges weakly on $[0,1]$ to $f(s)ds$. Its prefix sums are (OB17) up to one boundary atom; sandwich that atom between fixed nearby prefixes and use continuity of $\mathcal{J}$. The zero endpoint follows by first using a fixed positive prefix then tending to zero; the total mass follows from (OB16)'s single-window version. Step approximations then prove convergence against every continuous test function. This proves the correlated observed loss and its profile. It does not fix the consecutive complex phases in HG7, the separate kernel determinant, or the independent proper-source target minimum.

8 Nonvanishing of all original conductor corner coefficients

Retain the original ordered centered quartet, the original arithmetic unit, and the original evaluation matrix:

$$\begin{aligned} w_1 = x = \delta + i\gamma, \quad w_2 = \bar{x}, \quad w_3 = -\bar{x}, \quad w_4 = -x, \quad 0 < \delta < \frac{1}{2}, \quad \gamma > 2, \\ \beta = 2(\gamma^2 - \delta^2), \quad \eta = (\delta^2 + \gamma^2)^2, \quad h_c(w) = w^4 + \beta w^2 + \eta, \\ V_{\alpha, r+1} = w_\alpha^r \quad (1 \leq \alpha \leq 4, \quad 0 \leq r \leq 3), \quad U = \text{diag}(\mathbf{u}, \bar{\mathbf{u}}, \bar{\mathbf{u}}, \mathbf{u}), \quad \mathbf{u} \neq 0. \end{aligned} \quad (\text{OC50})$$

The four roots are distinct, so V is invertible. Put $\zeta_5 = e^{2\pi i/5}$, $D = \text{diag}(\zeta_5, \zeta_5^2, \zeta_5^3, \zeta_5^4)$ and $Q_0(t) = t_1 t_4 - t_2 t_3$. The original period factorization NC2–4, equivalently PCL1–7, is

$$\begin{aligned} H(\varpi) = D_\varpi \mathcal{R}(\varpi^{-1}) V^{-1} U, \quad \det \mathcal{R}(z) = 1, \quad \mathcal{R}(z)^{-1} = \text{adj } \mathcal{R}(z), \\ J := \mathcal{R}(0) = \begin{pmatrix} 1 & 0 & -\beta/3 & 0 \\ 0 & 1 & 0 & -2\beta/3 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix}. \end{aligned} \quad (\text{OC51})$$

Here D_ϖ is the complete original invertible diagonal ray matrix; it includes its exponential scalar, Gamma constants, and branch powers. The determinant identity in (OC51) is the exact PCL5 identity, not a determinant inferred from a truncation. Its full coefficient formula is recalled and bounded below.

Define the original conjugation matrices and their sixteen values by

$$\begin{aligned} L_j(z) = U^{-1} V \mathcal{R}(z)^{-1} D^{-j} \mathcal{R}(z) V^{-1} U, \\ q_{j, \alpha}(z) = Q_0(L_j(z) e_\alpha) \quad (1 \leq j, \alpha \leq 4). \end{aligned} \quad (\text{OC52})$$

The entries and all the displayed scalar functions are entire. In the original period coordinates, $H(\varpi)^{-1} D^{-j} H(\varpi) = L_j(\varpi^{-1})$ because D_ϖ commutes with D^{-j} . Thus (OC52) is an equality of the original maps with every ray factor cancelled against its actual inverse. It does not assert any invariance of a Hermitian output metric.

8.1 The exact primitive column and both original quadratic brackets

Write

$$\tau = \frac{\gamma^2 - \delta^2}{3} > 0, \quad d_h = 2\delta\gamma > 0, \quad \alpha = x^2 = -3\tau + id_h, \quad A = -\tau + id_h, \quad B = \tau + id_h. \quad (\text{OC53})$$

Polynomial division and evaluation at the four distinct roots give

$$\frac{h_c(w)}{w - x} = w^3 + xw^2 + (x^2 + \beta)w + x(x^2 + \beta), \quad V^{-1} e_1 = \frac{(x(x^2 + \beta), x^2 + \beta, x, 1)^\top}{h'_c(x)}.$$

Indeed this polynomial vanishes at the other three roots and has value $h'_c(x)$ at x . Since $\beta = 6\tau$, multiplication by the exact matrix J yields

$$J V^{-1} U e_1 = \frac{\mathbf{u}}{h'_c(x)} (xB, A, x, 1)^\top, \quad h'_c(x) = 4id_h x \neq 0. \quad (\text{OC54})$$

This proves the received equation (22) in its original coordinates.

For a vector Y , the coefficient polynomial of $J^{-1} Y$ is $P_Y(w) = (Y_1 + 2\tau Y_3) + (Y_2 + 4\tau Y_4)w + Y_3 w^2 + Y_4 w^3$. The products of its evaluations at $x, -x$ and at $\bar{x}, -\bar{x}$ give, without conjugating Y ,

$$\begin{aligned} \tilde{Q}(Y) &:= Q_0(U^{-1} V J^{-1} Y) \\ &= \mathbf{u}^{-2} \{ [Y_1 + AY_3]^2 - \alpha [Y_2 + BY_4]^2 \} \\ &\quad - \bar{\mathbf{u}}^{-2} \{ [Y_1 + \bar{A}Y_3]^2 - \bar{\alpha} [Y_2 + \bar{B}Y_4]^2 \}. \end{aligned} \quad (\text{OC55})$$

This is NC12 with its complete scalar, not the auxiliary real quadratic form obtained there by division by $2i$.

For $1 \leq j \leq 4$ set $\omega = \zeta_5^{-j} = e^{i\vartheta}$ and $r = \omega^2$. Rotating the numerator vector of (OC54) by D^{-j} gives $(\omega x B, \omega^2 A, \omega^3 x, \omega^4)^\top$. Its two brackets in (OC55) are exactly

$$\begin{aligned} F_\vartheta &= \alpha \omega^2 \{(B + Ar)^2 - r(A + Br)^2\} \\ &= \alpha \omega^2 (1 - r) \{B^2(1 + r)^2 - 4\tau^2 r\}, \\ G_\vartheta &= \omega^2 (1 - r)^2 \{Z - r\bar{Z}\}, \quad Z = \alpha B^2. \end{aligned} \quad (\text{OC56})$$

For the first factorization, subtract the two squares and use $(B - A)^2 = 4\tau^2$. For the second, use $\bar{A} = -B$, $\bar{B} = -A$ and $\bar{Z} = \bar{\alpha}A^2$. Homogeneity of degree two restores the square of the scalar in (OC54). Consequently

$$q_{j,1}(0) = \frac{F_\vartheta - (\mathbf{u}/\bar{\mathbf{u}})^2 G_\vartheta}{h'_c(x)^2}, \quad \omega = \zeta_5^{-j}. \quad (\text{OC57})$$

The phase in this formula is the original phase. No value for it has been selected.

8.2 Polynomial identity and a bound uniform in the full unit phase

Set $\epsilon = d_h/\tau$, $C = \cos^2 \vartheta$ and $S = \sin^2 \vartheta = 1 - C$. The identities $1 + r = 2\omega \cos \vartheta$ and $1 - r = -2i\omega \sin \vartheta$ give

$$\begin{aligned} Z/\tau^3 &= \epsilon^2 - 3 - i\epsilon(5 + \epsilon^2), \\ \frac{|F_\vartheta|^2}{16\tau^6|1 - r|^2} &= (9 + \epsilon^2) \{[(1 - \epsilon^2)C - 1]^2 + 4\epsilon^2 C^2\}, \\ \frac{|G_\vartheta|^2}{16\tau^6|1 - r|^2} &= S \{(3 - \epsilon^2) \sin \vartheta - \epsilon(5 + \epsilon^2) \cos \vartheta\}^2. \end{aligned} \quad (\text{OC58})$$

In particular the sign of the mixed term is fixed before the polynomial expansion. Expanding these two expressions, using $S = 1 - C$, and subtracting proves the exact received equation (24):

$$\begin{aligned} \frac{|F_\vartheta|^2 - |G_\vartheta|^2}{16\tau^6|1 - r|^2} &= 2\epsilon(5 + \epsilon^2)(3 - \epsilon^2)S \sin \vartheta \cos \vartheta \\ &\quad + \epsilon^2(7 - 21C + 50C^2) \\ &\quad + \epsilon^4(-1 - 6C + 20C^2) + \epsilon^6(2C^2 - C). \end{aligned} \quad (\text{OC59})$$

The accompanying script checks the identity over the rational polynomial ring, with zero residual, rather than at a list of numerical phases.

For a nonidentity fifth root, the two possible values of S are $(5 - \sqrt{5})/8$ and $(5 + \sqrt{5})/8$, and the two possible values of $(\sin \vartheta \cos \vartheta)^2$ are $(5 - \sqrt{5})/32$ and $(5 + \sqrt{5})/32$. Since $\sqrt{5} < 3$,

$$S \geq \frac{1}{4}, \quad |\sin \vartheta \cos \vartheta| \geq \frac{1}{4}, \quad |1 - r|^2 = 4S \geq 1.$$

Suppose $0 < \epsilon \leq 1/100$. The absolute value of the first term on the right of (OC59) is at least $2\epsilon \cdot 5 \cdot 2 \cdot (1/4) \cdot (1/4) = 5\epsilon/4$. The sum of the absolute values of the remaining terms is at most

$$78\epsilon^2 + 27\epsilon^4 + 3\epsilon^6 \leq 79\epsilon^2 \leq \frac{79}{100}\epsilon.$$

Here $0 \leq C \leq 1$, and $78 + 27/10000 + 3/10^8 < 79$. Therefore the absolute value of the whole right side of (OC59) is at least $23\epsilon/50$. Also $|\alpha| \leq 4\tau$, $|B|^2 \leq 2\tau^2$ and $|Z| \leq 8\tau^3$, so the formulas (OC56) give $|F_\vartheta| \leq 96\tau^3$ and $|G_\vartheta| \leq 64\tau^3$. For every $\lambda \in \mathbb{C}$ with $|\lambda| = 1$ it follows that

$$\begin{aligned} |F_\vartheta - \lambda G_\vartheta| &\geq ||F_\vartheta| - |G_\vartheta|| \\ &= \frac{||F_\vartheta|^2 - |G_\vartheta|^2|}{|F_\vartheta| + |G_\vartheta|} \geq \frac{23}{500}\tau^2 d_h. \end{aligned} \quad (\text{OC60})$$

The denominator is nonzero because its squared-modulus difference was just bounded strictly away from zero. Taking $\lambda = (\mathbf{u}/\bar{\mathbf{u}})^2$ proves the received equation (25), including every original unit phase.

The domain condition is explicit in the original root parameters:

$$\gamma \geq 1000\delta \implies 0 < \epsilon = \frac{6\delta\gamma}{\gamma^2 - \delta^2} \leq \frac{6000}{999999} < \frac{1}{100}. \quad (\text{OC61})$$

For an actual offcritical zeta quartet this domain follows from the finite-height theorem of Dave Platt and Tim Trudgian, *The Riemann hypothesis is true up to $3 \cdot 10^{12}$* , *Bulletin of the London Mathematical Society* 53 (2021), 792–797, doi:10.1112/blms.12460, arXiv:2004.09765. Theorem 1 [5] (source label **bank**) of the retained original TeX actually gives height 3 000 175 332 800; its abstract states the weaker $3 \cdot 10^{12}$ bound used here. The source’s “Theory and computation” section describes the interval-arithmetic sign counts and Turing-method completeness certificate. Thus any offcritical zero has $\gamma > 3 \cdot 10^{12}$, while $1000\delta < 500$. The algebraic proof (OC54)–(OC60) uses only its explicitly displayed parameter domain and is independent of that external finite-height proof.

8.3 Exact transport to the other twelve values

Let $E = \text{diag}(1, -1, 1, -1)$, let P_{opp} exchange indices 1, 4 and 2, 3, and let P_{conj} exchange 1, 2 and 3, 4. Put $T = JV^{-1}U$, so $L_j(0) = T^{-1}D^{-j}T$. Direct evaluation at the original roots gives

$$\begin{aligned} VE &= P_{\text{opp}}V, & JE &= EJ, & UP_{\text{opp}} &= P_{\text{opp}}U, & ET &= TP_{\text{opp}}, \\ \bar{V} &= P_{\text{conj}}V, & P_{\text{conj}}\bar{U} &= UP_{\text{conj}}, & \bar{T} &= TP_{\text{conj}}. \end{aligned} \quad (\text{OC62})$$

For example $VE = P_{\text{opp}}V$ says exactly that the r th monomial at $-w$ is $(-1)^r$ times its value at w . The second line also uses the real coefficients of J . Both E and D are diagonal, and $\bar{D} = D^{-1}$. These identities yield

$$\begin{aligned} P_{\text{opp}}L_j(0)P_{\text{opp}} &= L_j(0), & \overline{L_j(0)} &= P_{\text{conj}}L_{5-j}(0)P_{\text{conj}}, \\ Q_0(P_{\text{opp}}t) &= Q_0(t), & Q_0(P_{\text{conj}}t) &= -Q_0(t). \end{aligned} \quad (\text{OC63})$$

In the conjugation identity $5-j$ is again in $\{1, 2, 3, 4\}$. If $s_j = q_{j,1}(0)$, applying the two exact Q_0 identities to the corresponding columns gives the complete table

$$\begin{array}{c|cccc} \alpha & 1 & 2 & 3 & 4 \\ \hline q_{j,\alpha}(0) & s_j & -\overline{s_{5-j}} & -\overline{s_{5-j}} & s_j \end{array} \quad |q_{j,\alpha}(0)| \geq m_{\text{corner}} := \frac{23\tau^2 d_h}{500|h'_c(x)|^2} > 0. \quad (\text{OC64})$$

This proves the lower bound for all sixteen indices. In particular, conjugation transports a minus sign, although this sign does not affect the modulus bound. It must not be omitted from the exact morphism.

8.4 A closed finite majorant from the complete original coefficients

For clarity, the row index in the next calculation is $1 \leq b \leq 4$ and the centered-polynomial column index is $0 \leq r \leq 3$. The complete retained PCL2 formula is

$$\mathcal{R}_{br}(z) = \sum_{n \geq 0} \sum_{\substack{p, h \geq 0 \\ 2p+4h=5n+r+1-b}} \frac{\beta^p \eta^h}{3^p p! h!} (-1)^L 5^L (b/5)_L z^n, \quad L = p + h - n. \quad (\text{OC65})$$

The index r in $\mathcal{R}_{br}$ denotes its column labeled r , so in ordinary one-based matrix notation this is column $r+1$. Every contributing term has

$$n = \frac{2p+4h+b-r-1}{5}, \quad L = \frac{r+3p+h+1-b}{5}, \quad b/5 + L = \frac{r+3p+h+1}{5} > 0.$$

For contributing integer pairs $L \geq 0$, since a negative integer L would make $b+5L \leq -1$ although $r+3p+h+1 \geq 1$. Euler's defining integral for Gamma therefore rewrites the absolute value of the Pochhammer factor exactly. On $|z| = 2$, enlarging the resulting positive sum by dropping the congruence restriction gives

$$|\mathcal{R}_{br}(z)| \leq \frac{5^{(r+1-b)/5} 2^{(b-r-1)/5}}{\Gamma(b/5)} \int_0^\infty s^{(r+1)/5-1} \exp\{-s + A_2 s^{3/5} + B_2 s^{1/5}\} ds, \quad (\text{OC66})$$

$$A_2 = \frac{|\beta|}{3} 5^{3/5} 2^{2/5}, \quad B_2 = |\eta| 5^{1/5} 2^{4/5}.$$

This step preserves every original coefficient before estimating it. To justify the summation and integral, first perform it for finite positive sums, then use monotone convergence. The two unrestricted exponential series sum respectively to $\exp(A_2 s^{3/5})$ and $\exp(B_2 s^{1/5})$; the bound proved next shows the resulting integral finite. Thus there is no reliance on an unquantified tail convergence assertion in (OC66).

For $0 < p < 1$, $a \geq 0$ and $s \geq 0$, elementary maximization gives

$$as^p \leq \frac{s}{4} + (1-p)(4p)^{p/(1-p)} a^{1/(1-p)}.$$

For $a > 0$ this follows by differentiating $as^p - s/4$: its maximum is attained at $s = (4pa)^{1/(1-p)}$. For $a = 0$ it is immediate. Applying this separately with $p = 3/5$ and $p = 1/5$ proves

$$K_2 = \frac{2}{5} \left(\frac{12}{5}\right)^{3/2} A_2^{5/2} + \frac{4}{5} \left(\frac{4}{5}\right)^{1/4} B_2^{5/4}, \quad (\text{OC67})$$

$$M_{br} = 5^{(r+1-b)/5} 2^{b/5} \frac{\Gamma((r+1)/5)}{\Gamma(b/5)} e^{K_2}, \quad M = \max_{1 \leq b \leq 4, 0 \leq r \leq 3} M_{br},$$

$$|\mathcal{R}_{br}(z)| \leq M_{br} \leq M \quad (|z| = 2).$$

Indeed the integrand in (OC66) is at most $e^{K_2 s^{(r+1)/5-1}} e^{-s/2}$, whose integral equals $e^{K_2} 2^{(r+1)/5} \Gamma((r+1)/5)$. Every quantity in (OC67) is now a finite explicit function of the original β, η and positive Gamma constants.

Use the induced maximum-row-sum norm and put $\kappa = \|V\|_\infty \|V^{-1}\|_\infty$. There is also a direct formula retaining the original Vandermonde: with $\rho_0 = |x| = \sqrt{\delta^2 + \gamma^2} > 2$,

$$\|V\|_\infty = 1 + \rho_0 + \rho_0^2 + \rho_0^3, \quad \|V^{-1}\|_\infty = \frac{\rho_0^2}{d_h}, \quad \kappa = \frac{\rho_0^2(1 + \rho_0 + \rho_0^2 + \rho_0^3)}{d_h}. \quad (\text{OC68})$$

To prove the inverse norm formula, the cardinal column at any root w is $(w(w^2 + \beta), w^2 + \beta, w, 1)^T / h'_c(w)$. All four roots have modulus ρ_0 , $|w^2 + \beta| = \rho_0^2$ and $|h'_c(w)| = 4d_h \rho_0$. The four row sums are therefore $\rho_0^2/d_h, \rho_0/d_h, 1/d_h, 1/(d_h \rho_0)$, whose maximum is the first because $\rho_0 > 2$.

Each entry of $\text{adj } \mathcal{R}$ is a determinant of order three, with six summands. Equations (OC51) and (OC67) imply, on $|z| = 2$, $\|\mathcal{R}\|_\infty \leq 4M$ and $\|\mathcal{R}^{-1}\|_\infty \leq 24M^3$. The exact unit matrix has norms $\|U\|_\infty = |\mathbf{u}|$ and $\|U^{-1}\|_\infty = |\mathbf{u}|^{-1}$; these two moduli cancel in the estimate of the complete conjugation. Also $\|D^{-j}\|_\infty = 1$. Thus all sixteen functions satisfy the explicit circle bound

$$\|L_j(z)\|_\infty \leq 96\kappa M^4, \quad |q_{j,\alpha}(z)| \leq \mathcal{M}_2 := 18432\kappa^2 M^8 \quad (|z| = 2). \quad (\text{OC69})$$

The scalar inequality follows from $|t_1 t_4 - t_2 t_3| \leq 2\|t\|_\infty^2$ and $\|e_\alpha\|_\infty = 1$.

Writing $q_{j,\alpha}(z) = \sum_{n \geq 0} q_{j,\alpha,n} z^n$, Cauchy's coefficient formula on this specified circle gives

$$|q_{j,\alpha,n}| \leq \mathcal{M}_2 2^{-n}, \quad C_{\text{corner}} := \max_{j,\alpha} \sum_{n \geq 1} |q_{j,\alpha,n}| \leq \mathcal{M}_2, \quad (\text{OC70})$$

$$|q_{j,\alpha}(z) - q_{j,\alpha}(0)| \leq \mathcal{M}_2 |z| \quad (|z| \leq 1).$$

In the last step sum the coefficient bound and use $|z|^n \leq |z|$. This both establishes the incoming C_{corner} as an actual convergent coefficient sum and supplies a closed upper bound for it.

8.5 Original finite periods and all four corner coefficients

Let R_{adm} be the already specified original admissible radius. The original five-orbit requirement remains imposed there. Define

$$R_{\text{corner}}^{\text{explicit}} = \max \left\{ R_{\text{adm}}, 1, \frac{2\mathcal{M}_2}{m_{\text{corner}}} \right\}. \quad (\text{OC71})$$

For every original period with $|\varpi| \geq R_{\text{corner}}^{\text{explicit}}$, (OC64) and (OC70), at the actual $z = \varpi^{-1}$, prove

$$|q_{j,\alpha}(\varpi^{-1})| \geq m_{\text{corner}} - \frac{\mathcal{M}_2}{|\varpi|} \geq \frac{m_{\text{corner}}}{2} > 0 \quad (1 \leq j, \alpha \leq 4). \quad (\text{OC72})$$

The received threshold with $2C_{\text{corner}}/m_{\text{corner}}$ is valid and may be smaller; (OC71) makes its tail constant quantitatively explicit from the original period coefficients.

In the original affine Segre coordinates $t(Z, W) = (ZW, Z, W, 1)^\top$, the full biform is $A_z(Z, W) = \prod_{j=1}^4 Q_0(L_j(z)t(Z, W)) = \sum_{r,s=0}^8 a_{rs}(z)Z^rW^s$. For one quadratic factor its coefficients at $(2, 2), (2, 0), (0, 2), (0, 0)$ are respectively $Q_0(L_je_1), Q_0(L_je_2), Q_0(L_je_3), Q_0(L_je_4)$: in each case the indicated extremal degree can arise only by taking the square of that one Segre coordinate. In the four-factor product the same extremality forces that choice in every factor. Therefore

$$\begin{aligned} a_{88}(z) &= \prod_{j=1}^4 q_{j,1}(z), & a_{80}(z) &= \prod_{j=1}^4 q_{j,2}(z), \\ a_{08}(z) &= \prod_{j=1}^4 q_{j,3}(z), & a_{00}(z) &= \prod_{j=1}^4 q_{j,4}(z), \end{aligned} \quad (\text{OC73})$$

$$\min\{|a_{88}(\varpi^{-1})|, |a_{80}(\varpi^{-1})|, |a_{08}(\varpi^{-1})|, |a_{00}(\varpi^{-1})|\} \geq \left(\frac{m_{\text{corner}}}{2}\right)^4 > 0.$$

When the original notation writes $a_{88}(\varpi)$ for the coefficient at period ϖ , its value is exactly $a_{88}(z)$ at $z = \varpi^{-1}$ as used above. This proves the received equation (27) and the corresponding three remaining corner statements. Every original coefficient, scalar and unit phase remains in its defining product.

For completeness, on any already admitted domain $|\varpi| \geq R_* > 0$ the exceptional inverse periods for simultaneous nonvanishing are specified by the one original entire product

$$F_{\text{corner}}(z) = \prod_{j=1}^4 \prod_{\alpha=1}^4 q_{j,\alpha}(z), \quad \mathcal{Z}_* = \{z : |z| \leq R_*^{-1}, F_{\text{corner}}(z) = 0\}. \quad (\text{OC74})$$

By (OC64) this product is nonzero at zero. Its zeros are isolated; an infinite subset of the compact disk would have an accumulation point in its entire domain, contradicting the identity theorem. Thus $\mathcal{Z}_*$ is finite, and all four original corner coefficients are nonzero outside it. This is the *union* of the sixteen zero sets, equivalently the zero set of their product. The phrase “common zero set” in the incoming paragraph must not be interpreted as their intersection, since excluding only the intersection would not establish simultaneous nonvanishing. No lower bound is asserted at an exceptional period. The explicit large-period result (OC71)–(OC73) has already excluded every such zero in its domain.

The new conclusion supplied to the original observation calculation is the exact nonzero original terminal coefficient and its quantitative lower bound (OC73), with the period fixed while the cardinal degree grows. This proof does not invert the complete conductor or assign a value to the independent proper-source target minimum.

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